Who Is Really on the Frontier?
Confidence Sets for Cost-Accuracy Pareto Leaderboards

David Banahene[1] Jeffery Opoku[2,*]

[1]Robert Stempel College of Public Health and Social Work, Department of Biostatistics, Florida International University, Miami, FL 33199, USA
[2]School of Mathematical and Statistical Sciences, University of Texas Rio Grande Valley, Edinburg, TX 78539, USA

[*]Corresponding author: jeffery.opoku01@utrgv.edu

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Abstract

A leaderboard that reports cost beside accuracy is read through its Pareto frontier. This is the set of systems that no cheaper system beats. The frontier is drawn from point estimates on a sample of tasks, so it is an estimate, and no leaderboard reports how uncertain it is. We give a simultaneous confidence band for the frontier curve. We also give a confidence set for its members that contains the whole true frontier with probability at least $1 - \alpha$ in large samples. The band is proved for fixed or per-task costs. The set is proved for fixed costs, and for per-task costs when the true mean costs are distinct. A tie-robust set, a step-down on paired cost and accuracy comparisons, drops that condition at the price of a larger set. All guarantees are for the submitted systems and for tasks drawn independently from the benchmark's task process. They say nothing about tuning on public tasks. We check the methods on known truth in 18 scenarios and report where they fail. The failures include nearly tied costs, cheaper near-copies of other systems, and heavy-tailed costs. The band also fails when tasks are strongly clustered or cost logs are missing. From the public call logs of the Holistic Agent Leaderboard (HAL), we rebuild per-task costs for 324 runs on 10 benchmarks, and we reprice the run totals of 32 runs on an eleventh. HAL's cost totals double count nested trace spans on 202 of 332 runs. Counting each call once changes HAL's frontier on 2 benchmarks. Counting and pricing together change it on 8 of 11. With uncertainty added, the share of systems that cannot be excluded from the frontier runs from 42% to 100% across HAL boards. On a second leaderboard, HELM Capabilities, it runs from 36% to 61%, and its boards have 446 to 1000 tasks. Those shares assume that tasks are independent and that one run is a system's average. If the true standard errors were 1.5 times ours, the HAL standard sets would hold 316 of 356 systems instead of 280. Some pairs of an entrant and the holder it must beat are certified as new records on a random half of the tasks. On the other half, the mean lead is positive for 10 of 12 such pairs on HAL and 38 of 39 on HELM. The pairs are few, many are certified in only one or two splits, and the intervals are wide. The certificates and a winner's-curse correction for frontier members are in the appendices, each stated under its own null. The methods are in the open-source package `honestboard`.

1 Introduction

More and more leaderboards for AI agents report what a run costs beside how often it succeeds. The Holistic Agent Leaderboard (HAL) [15] does this for every benchmark it hosts. It draws the

cost-accuracy picture the way an economist would. A system is worth looking at only if no cheaper system does at least as well. The systems that pass this test form the Pareto frontier. HAL's own write-up stresses how few systems sit on it. Readers and the teams that build agents act on that picture. A system on the frontier is a reasonable one to consider at its price. A system below it usually is not.

The picture is drawn from point estimates. Each system's accuracy is its success rate on a sample of tasks: 33 on AssistantBench, 45 on CORE-Bench hard, 50 on tau-bench airline and on the SWE-bench Verified mini split. With samples that small, the rate has a standard error of several points. The frontier is a function of those rates, so it moves when the rates move. A system just below the frontier might sit on it under another draw of tasks. A system on it might not. HAL reports no uncertainty for its frontiers.

This paper supplies that uncertainty. It answers four questions.

1. How high, and how low, can the frontier curve really be? This curve is the best accuracy available at each cost. And which systems cannot be excluded from the frontier at a stated confidence level?

2. How wrong is the plain frontier when the truth is known, and where do the guarantees fail?

3. What is the cheapest system a reader cannot yet rule out as the best?

4. How many tasks would a board need before its frontier is resolved?

It also audits HAL with per-task costs rebuilt from its call logs. Two appendices give certificates for new frontier entrants and a winner's-curse correction for frontier members.

Contributions.

- A simultaneous confidence band for the frontier curve itself. It is valid with fixed or per-task costs, and with no condition on ties among the true mean costs (Section 3.2).

- A confidence set for the frontier's members. It contains the whole true frontier with probability at least $1 - \alpha$ in large samples, and it is proved for fixed costs and for per-task costs without ties. We also give a tie-robust version that drops the condition on ties, and exact finite-sample sets (Section 3.3).

- Resolution numbers, meaning the task count at which a board would start to resolve its own frontier. Also two descriptive summaries: a bootstrap frontier stability, and the cheapest unresolved competitor of the best (Sections 3.3 and 3.4).

- A known-truth validation in 18 scenarios. The scenarios include near-copy systems, outcome-dependent costs, nearly tied costs, heavy-tailed costs, missing cost logs and clustered tasks. The validation also has a stability check on held-out halves of real boards, and an account of where the guarantees fail (Section 5).

- An audit of HAL with consistent costing: each model call counted once, one price table, the double count quantified for every run, the effects of counting and of pricing on HAL's published frontiers reported separately, and the sensitivity of the audit to the costing rules, to clustered tasks and to run-to-run variation (Sections 4, 6 and 8).

- A second leaderboard, HELM Capabilities, audited with the same tools, and a fresh-task replication of the entrant certificates on both leaderboards (Section 7).

- In the appendices, two certificates for a new frontier entrant with chain-wise error control. One is a record claim, and one is a certificate of Pareto membership under a stated null. The appendices also give a proof that no certificate built on the entrant's strongest rival alone can hold its level, and the winner's-curse correction for frontier members.

What the paper does not claim. The tools are adapted. They are not invented from nothing. The set is Hsu's multiple-comparisons-with-the-best method. For each system, only the systems that cost no more count as rivals, and the margin is taken jointly. The exact sets are paired sign tests with Bonferroni. The certificates are the selective tests of a companion paper, with a different conditioning event. Our inference is conditional on the submitted systems. It takes the tasks as draws from the benchmark's task process. It does not account for tuning on public tasks, or for which systems were submitted (Section 2).

Related work. Earlier work has treated uncertainty in leaderboard rankings in several ways. These include confidence sets for ranks [22–24], confidence sets of the best [12, 13], the winner's curse of a selected system [1, 4], and standard errors for language-model evaluations [8, 20, 21]. The observed frontier is a free-disposal-hull estimator. Econometrics has developed inference for such frontiers, for sampled firms with exactly measured inputs and outputs. This includes the asymptotics of the free-disposal-hull estimator [25] and the bootstrap for nonparametric frontier models [27, 28]. Tests of stochastic dominance [18] ask a different question. Finance studies the sampling error of an efficient frontier [9]. Evolutionary optimisation studies the sampling error of Pareto fronts under noisy objectives [11]. Here the units are systems. Their coordinates, accuracy and cost, are both estimated from the same tasks, and the tasks are shared across units. Others have also studied how to identify a Pareto set from noisy observations when the analyst chooses what to sample next. Examples are Pareto-set identification with (ϵ, δ) guarantees from bandit feedback [3], active learning of the Pareto front [32], and multi-objective ranking and selection in simulation [2]. Cost has entered leaderboard design through HAL [15], through the argument that agent evaluations should report it [14], and through the cost-of-pass framework [10]. All of these work from point estimates. To our knowledge, no earlier work gives a published cost-accuracy frontier a simultaneous confidence band and a confidence set for its members. The literatures above are close, but a systematic comparison with them is beyond one paper.

Findings. HAL's cost totals double count nested trace spans on 202 of 332 runs. They also carry the prices of the month each run was made. Counting each call once changes the point-estimate frontier on 2 benchmarks. Counting and pricing together change it on 8 of 11. Under those changes, 10 of the 56 systems on the frontiers drawn from HAL's reported totals leave, and 18 others join. That is 28 changes in all, 24 of them through the pricing and 4 through the counting (Section 6). With uncertainty added, 64 systems sit on the observed frontiers. But the share that cannot be excluded runs from 42% on USACO to 100% on AssistantBench and SciCode. It tends to fall as the task count grows, though not monotonically. In the base simulations the set keeps the whole true frontier in 0.95 to 0.98 of boards against the nominal 0.95. With nearly tied true costs the standard set keeps it in only 0.26. With cheaper near-copies of other systems it keeps it in only 0.83. With heavy-tailed costs it keeps it in only 0.89. The tie-robust set keeps it in 1.00 and 1.00 of boards. In the nearly tied case it does so only by returning the whole board, because the data cannot resolve such a frontier (Sections 5 and 3.3). If the true standard errors were 1.5 times ours, the HAL standard sets would hold 316 of 356 systems instead of 280. Also, 5 of the 11 boards would have no system excluded at a factor of 2 (Section 6). On a second leaderboard, HELM

Symbol	Meaning
K, T	number of systems and number of tasks on a board
y_{kt}, $\bar{y}_k$, μ_k	score of system k on task t; its mean over the tasks; its expected score under the task process, which is the accuracy we want to know
c_{kt}, $\bar{c}_k$, γ_k	cost of system k on task t; its mean over the tasks; its expected cost
$A(x)$, $L(x)$, $U(x)$	the frontier curve (the best true accuracy among systems whose true cost is at most x), and the lower and upper edges of its band
α, B	error level, and number of bootstrap resamples
G_{ij}, $\widehat{\mathrm{se}}_{ij}$	how far system j sits above system i in standard-error units, and the paired standard error used
D_{ij}	the same for the cost difference $\bar{c}_i - \bar{c}_j$ (used only by the tie-robust set)
c_α	the margin: a bootstrap quantile in standard-error units
c_{band}	the bootstrap quantile used for the band (Section 3.2)
se_j, sg_j	the standard errors of $\bar{y}_j$ and of $\bar{c}_j$
$\widehat{F}_\alpha$, F	the confidence set for the frontier, and the set of systems it must contain (each proposition says exactly which)
f	the factor by which every standard error is multiplied in the sensitivity analysis of Section 6

Table 1: Notation used in Sections 3 to 6. Some letters are used again for a local purpose inside one proposition or appendix, for example the hypotheses A_{ij} and C_{ij} of Section 3.3. Each such use is defined where it occurs.

Capabilities, the share that cannot be excluded runs from 36% to 61%. Some pairs of an entrant and the holder it must beat are certified as new records on a random half of the tasks, in at least one split. On the other half, the mean lead is positive for 10 of 12 such pairs on HAL and 38 of 39 on HELM (Section 7). Many of these pairs are certified in only one or two splits.

2 Setting

A board has K systems evaluated on the same T tasks. System k scores $y_{kt} \in [0, 1]$ on task t. On most benchmarks this is a pass or a fail. On some it is a partial score. Its accuracy is the mean $\bar{y}_k = T^{-1} \sum_t y_{kt}$, which estimates $\mu_k = \mathbb{E}\, y_{kt}$. Its cost is either one number per system, c_k (the price of the run divided by T), or one number per task, c_{kt}. In the second case $\bar{c}_k$ is the mean, and it estimates $\gamma_k = \mathbb{E}\, c_{kt}$. On HAL the per-task costs come from the call logs (Section 4).

System j *dominates* system i when $c_j \le c_i$ and $\bar{y}_j \ge \bar{y}_i$ with at least one of the two strict. The *observed frontier* is the set of systems that no other system dominates. Two systems that are equal in both cost and accuracy do not dominate each other, so both can be on the frontier. This is the only tie convention we use, and we apply it throughout. The *true frontier* is the same set, with μ_k and γ_k in place of $\bar{y}_k$ and $\bar{c}_k$. A system is truly undominated when no other system dominates it in this sense.

What is estimated. We take the systems as given. Everything here is conditional on the K submitted systems. It says nothing about systems that were tried and not submitted, or tuned on the public tasks. The tasks are draws from the benchmark's task process. They are independent across tasks, but dependent across systems on the same task, because hard tasks are hard for everyone. The target is each system's expected accuracy μ_k on new tasks from that process. For a fixed set of tasks, the empirical frontier is exactly determined. The uncertainty we describe arises

only because the tasks are a sample. The target is a benchmark population, not a deployment population. A result about new kinds of tasks, other repositories or sites, or real use is not covered. Clustered tasks are where the result is most at risk (Section 8). Cost is the priced token usage of reproducing the run, in dollars per task under one price table. Its target is the expected value γ_k under the task process. So a frontier says which systems are cheapest to reproduce at a given accuracy, at the prices of the table. It does not say which were the best buys when they were run. The price table is a fixed list. A system's cost is the priced token count of its run, and this varies with the tasks drawn, so costs are random where prices are not. Three things are not modelled: tuning of a system on the public tasks, run-to-run variation of a system beyond independent task-level noise, and changes in prices over time.

3 Methods

3.1 The observed frontier and why it is not enough

The observed frontier is what every cost-aware leaderboard draws. It is a function of K estimated accuracies, each with a standard error of order $T^{-1/2}$. It is also a discontinuous function: moving one accuracy by a hair can add or remove a system. Section 5 measures how often it is wrong when the truth is known. In short, with 300 simulated boards of 20 systems on 50 tasks, 19% of the systems on the observed frontier are truly dominated, and 28% of the true frontier is missed.

3.2 A confidence band for the frontier curve

A cost-aware leaderboard draws a curve. At each cost x, the curve shows the best accuracy available from any system whose cost is at most x,

$$A(x) = \max_{j:\, \gamma_j \le x} \mu_j,$$

where γ_j is the true mean cost of system j. The curve is a staircase that steps up at the cost of each frontier member. The plain frontier is this curve drawn with the sample means $\bar{y}_j$ and $\bar{c}_j$ in place of μ_j and γ_j. A simultaneous band for A follows from a simultaneous band for all $2K$ means at once. Let se_j and sg_j be the standard errors of $\bar{y}_j$ and $\bar{c}_j$. The standard error is zero for a cost that is a fixed number. Resample the tasks, taking outcomes and costs together. For each resample, take the largest of $|\bar{y}_j^{(b)} - \bar{y}_j|/\mathrm{se}_j$ and $|\bar{c}_j^{(b)} - \bar{c}_j|/\mathrm{sg}_j$ over all systems. Let c_{band} be the $(1-\alpha)$ quantile of these largest values over the resamples. Each system then has an accuracy interval $[\bar{y}_j \mp c_{\mathrm{band}}\mathrm{se}_j]$ and a cost interval $[\bar{c}_j \mp c_{\mathrm{band}}\mathrm{sg}_j]$. The lower and upper edges of the band are

$$L(x) = \max_{j:\ \bar{c}_j + c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j - c_{\mathrm{band}}\mathrm{se}_j), \qquad U(x) = \max_{j:\ \bar{c}_j - c_{\mathrm{band}}\mathrm{sg}_j \le x} (\bar{y}_j + c_{\mathrm{band}}\mathrm{se}_j).$$

A system enters the lower edge only when its cost is surely within the budget x. It enters the upper edge when its cost possibly is. So neither edge relies on getting the cost order right.

Proposition 1 (band)**.** *Let the task vectors of outcomes and costs be i.i.d., with every random accuracy and cost having positive variance. Let K be fixed, and let each true accuracy μ_j be fixed in $(0,1)$ as $T \to \infty$. Then*

$$\mathbb{P}\big(L(x) \le A(x) \le U(x) \text{ for every } x \ge \min_j \gamma_j\big) \ \ge\ 1 - \alpha - o(1),$$

with costs fixed or measured per task, and with no condition on ties among the true mean costs. Below the cheapest true cost, $A(x)$ is a maximum over an empty set.

Proof. Let E be the event that every μ_j lies in its accuracy interval and every γ_j in its cost interval. If $\bar{c}_j + c_{\text{band}}\text{sg}_j \le x$ then on E $\gamma_j \le x$, so $\bar{y}_j - c_{\text{band}}\text{se}_j \le \mu_j \le A(x)$. So $L(x) \le A(x)$. If $\gamma_j \le x$ then on E $\bar{c}_j - c_{\text{band}}\text{sg}_j \le \gamma_j \le x$, so the maximizer j of μ_j over $\{\gamma_j \le x\}$ enters $U(x)$ with $\bar{y}_j + c_{\text{band}}\text{se}_j \ge \mu_j = A(x)$. So $U(x) \ge A(x)$. Both hold for every x on E. We use the multivariate central limit theorem and the consistency of the bootstrap for the mean vector [7]. Slutsky's theorem applies because the standard errors converge in probability. So the bootstrap law of the maximal studentised deviation converges to that of the maximum of the absolute values of a Gaussian vector whose coordinates have positive variance. Its distribution function is continuous and strictly increasing. So $\mathbb{P}(E) \ge 1 - \alpha - o(1)$. □

The proposition is asymptotic, with each μ_j fixed in $(0, 1)$. It does not protect the systems with few passes or few failures. A bootstrap-t interval covers too rarely for them. It shrinks to a single point when a system passes every task, or fails every task, in the sample. A system with true accuracy 0.02 on 50 tasks fails every task a third of the time. On a small board of four systems, the band built from bootstrap-t intervals alone covered the true curve only 62% of the time. So for pass/fail outcomes, a system with fewer than ten passes or ten failures gets a Clopper–Pearson interval at level $\alpha/(2K)$ in place of the bootstrap interval. The maximal deviation is taken over the other systems and all costs, at level $\alpha/2$. When every system is small, the maximal deviation is taken at level α, with Clopper–Pearson at α/K. When no system is small, there is no Clopper–Pearson part. The Clopper–Pearson half is exact, whatever the data select. Each interval covers with probability at least $1 - \alpha/(2K)$, whether or not its system is small. So the K of them fail together with probability at most $\alpha/2$. The maximal-deviation half is asymptotic. The proposition covers it when every system is regular, which means that the expected passes and failures both tend to infinity, so that no system is small with probability tending to one. For a sparse system, or one whose expected passes lie near ten, the data decide the switch between the two intervals. Then the total $1 - \alpha$ is a design target that the simulations support (Table 3). It is not a theorem. We designed this rule for the low-accuracy boards of HAL (AssistantBench, SciCode). Nothing in the proof uses the cost order. So the band does not need the cost order to be recoverable, but the frontier set of Section 3.3 does. The price is width. A system whose cost is uncertain is allowed into the upper edge long before it is allowed into the lower one. The band is a statement about the curve. The set is a statement about its members. The two use different statistics, and neither implies the other. The band is what to draw on a leaderboard page (Figure 1). Its width on HAL, as a median over the evaluated costs, runs from 6 points (ColBench frontend) to 62 points (CORE-Bench hard). In simulation it covers the true curve 0.94 to 1.00 of the time in 15 of the 18 scenarios. These include fixed costs, nearly tied costs and the low-accuracy board (Tables 3 and 4). It falls short in two of the last three rows of Table 4. The coverage is 0.89 when failed tasks have no logged cost. It is 0.71 when tasks are strongly clustered, and resampling the groups brings it back to 0.97 (Section 5.1). On most boards the coverage is above the nominal level. This is because the band takes one maximal deviation over all $2K$ means, for a curve that depends on one maximum at each cost. So the band is conservative by construction. A first version of the band trusted the sample cost order. It covered the true curve only 18 to 62% of the time on the simulated boards of Tables 3 and 4. This is why the construction above lets a system into an edge only on the strength of its cost interval.

3.3 A confidence set for the frontier

Call j a *rival* of i when $c_j \le c_i$ and $j \ne i$. A rival is any system that could dominate i on cost. For each pair, let $\widehat{\text{se}}_{ij}$ be the paired standard error of $\bar{y}_j - \bar{y}_i$. We compute it from the task-level

covariance of the two outcome series. Let

$$G_{ij} = \frac{\bar{y}_j - \bar{y}_i}{\widehat{\mathrm{se}}_{ij}}$$

be how far j sits above i in standard-error units. Two systems with identical outcomes on every task have a zero paired standard error and a zero gap. We set the standard error to 10^{-12}, so that $G_{ij} = 0$ and the pair never excludes a system. System i is on the observed frontier exactly when $G_{ij} < 0$ for every rival (with the tie conventions). The confidence set keeps i unless some rival beats it by more than a simultaneous margin:

$$\widehat{F}_\alpha = \Big\{ i : \max_{j \text{ rival of } i} G_{ij} \le c_\alpha \Big\}.$$

The margin is a bootstrap quantile. Resample the T tasks with replacement B times. In resample b, write $\bar{y}_k^{(b)}$ for the accuracies. When costs are per task, write $\bar{c}_k^{(b)}$ for the costs, so that the rival sets are resampled too. Let

$$M_i^{(b)} = \max_{j \text{ rival of } i \text{ in resample } b} \frac{(\bar{y}_j^{(b)} - \bar{y}_i^{(b)}) - (\bar{y}_j - \bar{y}_i)}{\widehat{\mathrm{se}}_{ij}},$$

which is the largest centred, studentised deviation of any rival from i. Let $M^{(b)} = \max_i M_i^{(b)}$ be the largest such deviation on the board. The margin is the $(1-\alpha)$ quantile of $M^{(b)}$ over resamples, $c_\alpha = \max(0,\, q_{1-\alpha}(M))$. This is Hsu's multiple-comparisons-with-the-best method [13], as used for the confidence set of the best in Banahene and Opoku [5]. Here the rival set of each system is cut down to the systems that cost no more, and the margin is taken jointly over systems. Two properties are immediate. First, $\widehat{F}_\alpha$ always contains the observed frontier, because a frontier member has every $G_{ij} < 0 \le c_\alpha$. Second, it shrinks toward the observed frontier as T grows, because the margin is in standard-error units.

Proposition 2 (the set contains the true frontier)**.** *Fix the costs. Let the task vectors be independent and identically distributed with finite variances, with the population variance of $y_{jt} - y_{it}$ positive for every pair compared, and let K be fixed as $T \to \infty$. Let F be the set of systems that no rival beats in true accuracy, the i with $\mu_j \le \mu_i$ for every rival j. It contains the true frontier of Section 2, which also drops a system tied in true accuracy with a strictly cheaper one. Then*

$$\mathbb{P}(F \subseteq \widehat{F}_\alpha) \;\ge\; 1 - \alpha - o(1).$$

Proof. For $i \in F$ and any rival j, $\mu_j - \mu_i \le 0$, so $G_{ij} \le Z_{ij} := \{(\bar{y}_j - \bar{y}_i) - (\mu_j - \mu_i)\}/\widehat{\mathrm{se}}_{ij}$. So $\max_{i \in F} \max_j G_{ij} \le \max_{i \in F} \max_j Z_{ij} \le \max_i \max_j Z_{ij} =: Z$. Each maximum is over the system's rivals, and these sets do not change with the data when costs are fixed. By the multivariate central limit theorem, the centred, scaled mean vector is asymptotically Gaussian. The nonparametric bootstrap of the mean vector is consistent [7]. Both Z and M divide by the sample standard errors $\widehat{\mathrm{se}}_{ij}$, which converge in probability to their population values. So by Slutsky's theorem and the continuous mapping theorem, Z and the bootstrap M share the limit law of the maximum of the coordinates of a Gaussian vector with positive coordinate variances. The distribution function of this law is continuous and strictly increasing. So c_α converges to that law's $(1-\alpha)$ quantile, and $\mathbb{P}(Z > c_\alpha) \to \alpha$. So $\mathbb{P}(F \not\subseteq \widehat{F}_\alpha) = \mathbb{P}(\max_{i \in F} \max_j G_{ij} > c_\alpha) \le \mathbb{P}(Z > c_\alpha) \le \alpha + o(1)$. □

Remark 1. *This is a confidence set for the frontier as a set. With probability about $1-\alpha$, every truly undominated system is kept at once. Two smaller sets with weaker, per-system guarantees*

come from the same resamples. The first uses the largest per-system quantile, $\max_i q_{1-\alpha}(M_i)$*, as the margin. The second uses each system's own quantile* $q_{1-\alpha}(M_i)$*. Each keeps a given truly undominated system with probability about* $1-\alpha$*. Section 5 reports all three against the alternatives a practitioner might reach for instead. On HAL the three sets hold 280, 259 and 248 systems. We call this joint set the frontier set. When we compare it with the tie-robust set, we also call it the standard set.*

Proposition 3 (per-task costs)**.** *Let costs be measured per task, with the task vectors of outcomes and costs i.i.d., every accuracy and cost having positive variance and* $y_{jt} - y_{it}$ *having positive variance for every pair. Let the true mean costs* $\gamma_1, \ldots, \gamma_K$ *be distinct. Let* F *be defined from the true accuracies and true mean costs. Then the frontier set built with per-task costs resampled jointly with the outcomes satisfies* $\mathbb{P}(F \subseteq \widehat{F}_\alpha) \geq 1 - \alpha - o(1)$*. (The band of Proposition 1 needs no such condition.)*

Proof. With distinct true mean costs, the sample mean costs recover the true cost order with probability tending to one. So do the resampled mean costs, within every resample except a vanishing share, since each is $\sqrt{T}$-consistent for its target. On that event every rival set, real and resampled, equals the true rival set. The statistics then coincide with those of the fixed-cost construction with the true rival sets, so Proposition 2 applies. Ties in true mean cost are the case the proof excludes. There, a rival set stays random in the limit. Proposition 4 below removes the condition. □

Remark 2. *On HAL the per-task and fixed-cost versions give identical sets on 10 of the 10 boards with per-task costs. Cost differences between neighbouring systems are mostly many times their standard error, so on these boards cost uncertainty does not move the set. That is a fact about HAL's cost spread, not a general property. The next paragraph shows what happens without it. Reruns of one configuration have the same true cost. This breaks the distinctness condition of Proposition 3 for those pairs, but in the harmless direction. Whether the sample order puts a twin among a system's rivals or not, the twin is a true rival. So a wrong order can only withhold evidence from the set.*

When true costs tie. Proposition 3 needs distinct true mean costs, and near ties are what break it. Take a system that is truly cheaper than a rival by one per cent, with the rival truly more accurate. The system is on the true frontier, because the rival does not cost less. But when the rival's sample cost happens to come out lower, the standard set treats the rival as a rival. It may then exclude the system on its evidence. In the nearly tied scenario of Section 5.1, true mean costs are a few per cent apart and per-task costs have a coefficient of variation of 0.8. There the standard set contains the whole true frontier in only 0.26 of simulated boards (Table 4), against the nominal 0.95. The same failure arises when a near-copy of another system costs a few per cent less than the original (0.83). The remedy has two parts. We exclude a system only when another system is shown to be strictly cheaper and strictly more accurate. And we use one critical value for both kinds of evidence.

The procedure is a step-down in the sense of Romano and Wolf [26]. It works on two families of one-sided hypotheses, one pair for every ordered pair of systems. C_{ij} says that j is not strictly cheaper than i, $\gamma_j \geq \gamma_i$. A_{ij} says that j is not more accurate than i, $\mu_j \leq \mu_i$. Let $D_{ij} = (\bar{c}_i - \bar{c}_j)/\widehat{\mathrm{se}}^c_{ij}$ be the studentised mean cost difference. Here $\widehat{\mathrm{se}}^c_{ij}$ is the paired standard error of the cost difference. Costs move together across tasks, so it is much smaller than the sum of the two unpaired standard errors. G_{ij} is as before. Start with no hypothesis rejected. At each step, let c be the $(1-\alpha)$ bootstrap quantile of the largest centred, studentised deviation over the hypotheses that still stand. Reject

every standing C_{ij} with $D_{ij} > c$ and every standing A_{ij} with $G_{ij} > c$, recompute c, and repeat until nothing changes. The tie-robust set excludes i when some j has both C_{ij} and A_{ij} rejected.

Proposition 4 (tie-robust set)**.** *Let costs be measured per task, with the task vectors of outcomes and costs i.i.d., every accuracy and cost having positive variance and the paired differences of accuracies and of costs having positive variance for every pair. There is no condition on ties among the true mean costs. Let F be the set of systems i with $\mu_j \le \mu_i$ for every $j \ne i$ with $\gamma_j \le \gamma_i$. The tie-robust set contains F with probability at least $1 - \alpha - o(1)$, and it contains the observed frontier.*

Proof. Let $\mathcal{N}$ be the set of true hypotheses, and let W be the event that none of them is rejected. A true hypothesis has a studentised statistic no larger than its centred version, so rejecting it needs its centred deviation to exceed the current c. Let k be the first step at which a true hypothesis would be rejected. Every hypothesis rejected before step k was false, so the hypotheses standing at step k include all of $\mathcal{N}$. The critical value c_k is then at least the $(1 - \alpha)$ bootstrap quantile of the largest centred deviation over $\mathcal{N}$. By the multivariate central limit theorem and the consistency of the bootstrap for the mean vector [7], that quantile converges to the $(1 - \alpha)$ quantile of the maximum of a Gaussian vector with positive coordinate variances. The distribution function of that maximum is continuous and strictly increasing. So the probability that a true hypothesis is rejected at step k is at most $\alpha + o(1)$, and $\mathbb{P}(W) \ge 1 - \alpha - o(1)$. On W, an excluded i has some j with C_{ij} and A_{ij} both false, that is $\gamma_j < \gamma_i$ and $\mu_j > \mu_i$, so $i \notin F$. If i is on the observed frontier and both C_{ij} and A_{ij} were rejected, then $\bar{c}_j < \bar{c}_i$ and $\bar{y}_j > \bar{y}_i$, so j dominates i in the sample. This is impossible. □

The proof uses nothing about how the cost and accuracy sampling errors depend on each other. We also checked the set in a demanding case by simulation. Every system has the same true accuracy, so every system is in F and any exclusion is an error. True costs are spread so that their gaps are of the order of the critical value. In 3 of the 4 settings, failed tasks cost three times as much as passed ones. The set kept all of F in 0.98 to 0.99 of boards, against the nominal 0.95 (supplement S12).

A *gated* variant tests A_{ij} only once C_{ij} has been rejected. The gate means that fewer hypotheses are tested at the start, so the sets are smaller, at most 34% above the standard set, against at most 37% for the set above. We do not use it, because the argument above does not cover it. With the gate, the number of hypotheses at risk grows as accuracy hypotheses are exposed, so the critical value is not monotone across steps. Also, which hypotheses are exposed depends on the cost differences, and these are correlated with the accuracy differences when failed tasks cost more. In simulation it kept the whole true frontier in at least 0.987 of boards in every one of the 18 scenarios, and in 0.97 to 0.98 in the demanding case. So the proof, not the evidence, is what is missing.

The price is size. When costs are tied more closely than their noise, no system is shown to be cheaper than another. Nothing is excluded, and the set is the whole board. That is the honest answer: the data cannot say which systems are cheaper, so they cannot resolve the frontier. The set does not shrink to the true frontier when true costs are exactly tied. A truly dominated system with an exactly tied dominating twin is kept at any T, because the twin is almost never shown to be cheaper. Where costs do differ, the set is larger than the standard set by up to 37% in simulation. That excess is the price of not knowing the cost order for the pairs whose order the data do not resolve. These pairs cannot be used to exclude a system, while the standard set uses them by guessing the order. The interval construction of honestboard 0.5.1 is more conservative still, up to 41% above the standard set. On HAL the tie-robust set holds 299 systems against 280 for the standard set. It differs from the standard set on 5 of the 10 boards with per-task costs

(Table 6). The paired cost comparison resolves the order of 0.74 to 0.98 of the pairs of systems on the 10 boards with per-task costs, against 0.82 in the base simulations and 0.01 in the nearly tied scenario (supplement S7). So HAL's boards are not in the nearly tied regime. On CORE-Bench hard the set holds all 60 systems. This is a caution: the cost order there is only partly resolved.

An exact set for any number of tasks. Proposition 2 is asymptotic. A finite-sample companion for fixed costs keeps i unless some rival j beats it in a one-sided exact paired test at level α/m_i. Here m_i is the number of i's rivals. The test is the sign test on the discordant tasks for pass/fail outcomes, and a sign-flip test for rates. The sign-flip test also needs the paired differences to be symmetric under the null. A truly undominated system has no rival that is truly better. So under the task-wise null, each rival's test rejects with probability at most α/m_i. By Bonferroni over its rivals, the system is kept with probability at least $1-\alpha$, for any T. Dividing by the number of rival pairs on the whole board instead gives a joint version. It contains the whole true frontier with probability at least $1-\alpha$ for any T. With per-task costs, the rival sets come from the sample cost order, and a rival may truly cost more. The finite-sample guarantee then does not extend, and the exact sets are only a comparison, not a guarantee. The price is power. The exact sets are the bootstrap set's conservative cousins. Like for like, on HAL the exact per-system set holds 269 systems against 248 for the per-system set of the bootstrap, and the exact joint set holds 291 against 280 for the joint set (Table 8).

What it would take to resolve a board. A system that is in the set but off the plain frontier is one whose strongest cheaper rival leads it by G standard errors, with $G \leq c_\alpha$. The lead in standard-error units grows like $\sqrt{T}$. So with T' tasks it would be $G\sqrt{T'/T}$, and the system leaves the set once $T' > T(c_\alpha/G)^2$. Section 6 reports two summaries of this for each board. The first is the task count at which half of the systems that are in the set but off the plain frontier would be excluded. The second is the task count at which a system trailing a cheaper rival by five accuracy points would be excluded, given the board's typical paired noise. Neither is a sample-size formula for a new benchmark, because the gaps on a future board are unknown. They say how far the present boards are from resolving the frontiers they draw.

3.4 Two descriptive summaries

Bootstrap frontier stability. The bootstrap frontier stability of system i is the share of task resamples in which it is on the observed frontier. Costs are resampled too when they are per task. This share describes how stable the system's membership is under the task process. It is not a probability that the system is truly undominated, and it is not calibrated. With 20 systems on 50 tasks it averages 0.63 over truly undominated systems and 0.10 over truly dominated ones (Table 3). We report it only as a stability measure.

The cheapest unresolved competitor of the best. The confidence set of the best [13] is the set of systems that we cannot rule out as the most accurate. It makes no reference to cost. It is the construction of Section 3.3 with every other system as a rival. Its cheapest member answers a question a reader asks: which is the cheapest system the data cannot yet rule out as the best? We call this member the cheapest unresolved competitor of the best. It is not an equivalence claim. When the board is small, a system can be materially worse and still be unresolved. Not rejecting a system is not evidence that it is equal. Its coverage follows from the coverage of the set. The true best is in the set with probability at least $1-\alpha$. When it is, the cheapest member costs no more than the true best on the observed costs. For the true costs to be ordered like the observed

costs, the cost order must be recoverable. The proof does not supply this. In simulation, the true best is in the set in 0.97 to 1.00 of boards over all 18 scenarios. Table 4 shows nine of them. The competitor's true cost is no more than the true best's in 0.97 to 1.00 of boards. The exception is when true costs are nearly tied, where it is 0.87 (Table 4).

3.5 Extensions: new entrants and selected winners

A live board changes when new submissions arrive. Also, a system reported as on the frontier was selected for being there. Two results address these facts. They build on the selective inference of the companion paper [5]. They are in the appendices, so the main text stays about the frontier as an estimated object.

Certifying a frontier entrant (Appendix A). A new system is on the observed frontier exactly when it beats the most accurate system that is cheaper than it. So announcing an entrant is a record claim against one holder. The exact selective test of the companion paper certifies that record claim. It controls the chain-wise error over the board's history. The record claim is not a claim of Pareto membership. We prove that no certificate built on the entrant and its strongest rival alone can hold its level against the composite null that some cheaper system dominates the entrant. We give certificates of Pareto membership under a conditional null. One form uses a normal approximation. The other form is exact for pass/fail outcomes. We walked HAL's boards in submission order. 69 systems were testable entrants, meaning they beat every cheaper system on arrival. Another 43 were simply the cheapest so far, with no rival to beat. Of the entrants, 6 can be certified as record claims, and 8 as Pareto members in the normal form. Both counts use alpha-spending over each board's chain. Section S2 of the supplement gives the counts board by board. It also shows how the counts depend on how submissions are batched, and it lists every certification step.

The frontier winner's curse (Appendix B). The selection that put a system on the frontier inflates its reported accuracy. The corrections of the companion paper apply here, with the pool of cheaper systems in place of the board. On HAL they take 1.7 points off a member's reported accuracy on average, and up to 9.4 in the largest case.

3.6 What is proved and what is simulated

Table 2 lists, for each tool, what is proved and under which conditions. It also lists what rests on simulation or held-out data.

4 Data: HAL with per-task costs rebuilt

For every run it hosts, HAL publishes an encrypted trace archive in its Hugging Face dataset (`agent-evals/hal_traces`). The dataset holds about 113 GB across 380 archives at the time of the audit. Each archive holds the run's configuration, its per-task outcomes, and its total cost as HAL computed it. It also holds the full call log. The log lists every model call the agent made, tagged with the task it served and the tokens it used. The decryption key is the public key in HAL's own harness. We read only the results member of each archive, by HTTP range requests. A results member is a few hundred kilobytes to a few hundred megabytes. We reduced each one on the fly to a compact record of outcomes, token counts and costs per task. We skipped 5 archives,

Tool	Proved	Checked by simulation or held-out data
Frontier band	Covers the whole frontier curve with probability $\geq 1-\alpha$, asymptotically, for fixed or per-task costs, with no condition on ties (Prop. 1). The Clopper–Pearson part, for systems with few passes or failures, is exact. Approximation only: the rule that switches a system to Clopper–Pearson depends on the data. So for sparse systems $1-\alpha$ is a design target, not a theorem	Coverage of the true curve of at least 0.94 in 16 of 18 scenarios. It falls short with missing cost logs, and with strongly clustered tasks unless groups are resampled (Tables 3 and 4)
Frontier set	Contains the whole true frontier with probability $\geq 1-\alpha$, asymptotically, for fixed costs (Prop. 2) and for per-task costs without ties (Prop. 3)	Retention against six alternative sets (supplement S11). Eight stress tests (Table 4): the set holds near 0.95 in five, and falls short with nearly tied costs, cheaper near-copies and heavy-tailed costs. Stability on held-out halves (Table 5)
Tie-robust set	A step-down on paired cost and accuracy comparisons. It contains the whole true frontier with probability $\geq 1-\alpha$, asymptotically, with no condition on ties or on how the cost errors and accuracy errors depend on each other (Prop. 4). A gated variant is supported by simulation only	Retention and size in the eight stress tests, including nearly tied costs, cheaper near-copies and heavy-tailed costs (Table 4). A hard case (supplement S12). Size on HAL (Table 6) and HELM (Table 10)
Exact frontier sets	With fixed costs, and under the task-wise null of Section 3.3, per-system retention is $\geq 1-\alpha$ for any T. The joint version contains the true frontier with probability $\geq 1-\alpha$ for any T (Section 3.3). There is no guarantee with per-task costs	Sizes compared with the bootstrap sets, on known truth and on HAL
Bootstrap frontier stability	Nothing. It is a description	Mean over truly undominated and truly dominated systems
Cheapest unresolved competitor	Follows from the coverage of the best set [13], on observed costs	Share of boards on which the true best is in the best set and the competitor costs no more (Table 4)
Entrant certificates (Appendix A)	Record claim: exact selective test under the conditional task-wise null (Prop. 5). Pareto certificate: composite null, with a normal approximation that uses a plug-in covariance and a lattice correction, so approximation only (Prop. 6). It is exact for pass/fail outcomes under a conditional null (Prop. 7). No pairwise certificate holds its level (Prop. 8). Chain-wise error control, with costs fixed in advance (Cor. 1)	Level and power with near-copy and independent rivals (Table 12). Sensitivity to batching. Replication of certified entrants on fresh tasks (Table 11)
Winner's curse (Appendix B)	The conditional estimate is exact under normality for a fixed pool [16]	Bias and coverage on known truth (Table 13) and on held-out halves (supplement S3)

Table 2: What each tool guarantees. Every guarantee holds under the estimand of Section 2. This means tasks are drawn independently from the benchmark's task process, one run is taken as a system's average, inference is conditional on the submitted systems, and there is no protection against tuning on public tasks. "Approximation only" marks a statement that rests on a normal approximation or a data-driven rule, not on a theorem.

because their results member was too large to decrypt in memory on our machine. The largest of these was 12.8 GB. We read the other 375.

Per-task cost. A task's cost is the priced token usage of the model calls logged under it. We price the usage with HAL's own formula: fresh input tokens at the model's input price, cache reads and writes at the model's cache price, and output tokens at the output price. We use one price table for every run, which is HAL's own as read on 13 October 2026. We count each model call once. The last point matters. HAL's reported total sums the usage block of every span in the trace. An agent that calls a provider through a wrapper library logs two spans for one call: the wrapper, and the provider call inside it. Both carry the same usage block. This effect was reported on the dataset's discussion board in September 2026, and the HAL team confirmed it [19]. HAL's pages now carry a caveat, but the totals stand. Of the 332 runs we read with usage, 141 have a reported token total of exactly twice the once-counted total. 44 fall between once and twice, 17 fall above twice, and 130 are unaffected. Over all runs, the reported totals overstate token use by 49%. The doubling depends on the agent's software stack, not on the model. So it can change the cost order of agents on the same board. On the boards where it changes HAL's frontier, it moves 4 systems (Table 7). We drop the duplicate spans. A duplicate is a call whose parent carries an identical usage block, or whose children's usage sums to its own. The supplement lists the ratio for every run. The rule leaves some structures unclear, so a different rule could remove more or fewer spans. A rule that removes some, but not all, of the spans we remove gives costs between HAL's reported totals and ours. Table 7 draws the plain frontier under both ends. No person has yet checked a stratified sample of traces against the rule by hand. As a check, we summed our per-task costs and compared them with HAL's reported total. They match to the cent on runs without nested spans that were made after the last price change of the models they used. They equal exactly half of HAL's total on runs where every provider call was wrapped. Over all 128 kept runs with no nested spans, the recount matches HAL's total within a cent plus half a per cent on 115. The other 13 differ because HAL's total embeds the prices in force when the run was made. For example, an o3 run from May 2025 reports several times what the current table gives for the same tokens, because o3's price fell by four fifths in June 2025. An early CORE-Bench run priced cached tokens at the full input rate. A frontier compares systems with one another, so it needs one price list and one way of counting. We reprice and recount everything. We report HAL's figure beside ours in the released tables. The supplement lists every archive with HAL's total, the ratio of token totals, our recount and the run's status. The two effects compound. Over the 318 runs costing at least a dollar, HAL's reported totals sum to $44,679, against $26,782 once counted and repriced. The median run is overstated by a factor of 2.00. The overstatement is more than half for 179 runs, and the largest gap is a factor of 31.

Outcomes and task sets. Outcomes come from the run's lists of successful and failed tasks, or from its per-task evaluation results. A benchmark's task set is the tasks with an outcome in at least half of its runs. A task without a logged outcome in a run counts as a fail. This touches 9 of the 356 kept runs, and each of them misses at most 5% of its tasks. Reruns of one configuration, and HAL's test uploads of the browser agents, are separate systems, as HAL lists them. They are each other's rivals at similar cost. Their costs are a median factor of 1.09 apart (Section 8). They add to the dominated-on-arrival count of supplement S2. On ColBench backend programming, HAL's accuracy is the share of tasks scored exactly one, and we use that. On ColBench frontend design, HAL's metric is a continuous similarity score in $[0, 1]$. We keep it, so "accuracy" there is a mean score.

Exclusions. Of the 375 run records, 356 enter the audit. We dropped 11 runs because they carried outcomes for fewer than 95% of the benchmark's tasks. We dropped 8 runs because their call logs carried no model call on more than half of their tasks. An example is runs of Claude Code, which does not log through HAL's tracer. For these runs the cost is unknown, not zero. A system with an unknown cost cannot be placed on a cost-accuracy frontier, so dropping it is the only honest option. The supplement lists every dropped and skipped run. Some tasks of a kept run have an outcome but no logged model call. This happens with a logging gap, or when a run failed before its first call. We price these tasks at the run's mean cost over its costed tasks, not at zero, because a zero would push the run toward the frontier. Of the kept runs, 47 have at least one such task, and 7 have more than a tenth. Of all system-task pairs, 1.5% are imputed. The supplement gives the count per system. On Online-Mind2Web, the browser agents' logs carry a call record per step but no token usage on any call. So no per-task cost exists for any run there, and there are no per-call usage blocks to double count. The run totals, which HAL computed from the stored run-level usage, reprice exactly under the current table. The only exception is two o3 runs, for the reason above. We use each run's repriced total, spread evenly over its tasks. So on that board, cost is a fixed number per system and is not resampled. Wherever the board appears in a table or figure, it is marked with a dagger. The dagger means that cost is a fixed number per system.

Why only HAL. Terminal-Bench 2 publishes cost for three of 244 job files, and it has no per-task results. SWE-bench Verified's cost metadata covers 47 submissions, without per-task results. Neither supports paired inference. So the audit rests on HAL's 11 benchmarks. Every kept run on 10 of them has per-task outcomes and per-task costs.

5 Validation on known truth

We simulated boards with $K \in \{8, 20\}$ systems and $T \in \{50, 150, 500\}$ tasks. True accuracies and mean costs were spread so that several systems are truly undominated. Task difficulty was shared across systems, so outcomes are correlated within a task. Per-task costs vary around each system's mean. On each of 300 boards per scenario, we computed the observed frontier and the frontier set from the sample and compared them with the true frontier. A share near 0.95 that is estimated from 300 boards has a Monte Carlo standard error of at most 0.013. So shares from about 0.93 up are consistent with the nominal level. Table 3 has the results for ten scenarios: the six board shapes of the base design and four other scenarios, described below. Every truly undominated system was kept at once on 0.95 to 1.00 of boards (0.95 to 0.98 for the six base shapes). A given truly undominated system was kept 0.991 to 1.000 of the time. Over the same boards, the plain frontier wrongly included 5 to 29% of its members and missed 6 to 41% of the true frontier. The price of the guarantee is size. With 20 systems, the set held 16.9 systems on average against a true frontier of 5.8 at $T = 50$, 13.4 against 5.6 at $T = 150$, and 9.9 against 5.7 at $T = 500$. The set does not become informative until a board has a few hundred tasks. This is the central practical message of the paper. Four of the ten scenarios probe the assumptions of the base design. In the first, costs are fixed at their means, which is the case Proposition 2 covers. In the second, the link between cost and ability is weaker. The third is a crowded board of 60 systems on 45 tasks, which has the shape of CORE-Bench hard. The fourth is a board of low accuracies: 20 systems on 65 tasks, most of them passing only a handful of tasks. It has the shape of SciCode, and there the band needs its exact intervals. With fixed costs the set holds 13.6 systems, against 13.4 with per-task costs on the same board shape. On the crowded board it holds 49.0 of 60 systems against a true frontier of 8.3, and 29% of the plain frontier is truly dominated. This is the picture the audit

				Frontier set			Band	Plain frontier		Stability	
Systems	Tasks	Costs	Corr.	kept, each	kept, all	size (true)	covers	falsely on	missed	true	dominated
8	50	per task	0.6	0.994	0.98 (0.009)	7.1 (4.0)	0.98 (0.008)	12%	17%	0.75	0.15
8	150	per task	0.6	0.991	0.97 (0.010)	6.2 (3.9)	0.98 (0.008)	8%	11%	0.84	0.11
8	500	per task	0.6	0.993	0.97 (0.009)	5.2 (3.9)	0.98 (0.008)	5%	6%	0.91	0.07
20	50	per task	0.6	0.996	0.98 (0.008)	16.9 (5.8)	1.00 (0.000)	19%	28%	0.63	0.10
20	150	per task	0.6	0.992	0.95 (0.012)	13.4 (5.6)	0.98 (0.007)	12%	15%	0.78	0.07
20	500	per task	0.6	0.994	0.96 (0.011)	9.9 (5.7)	0.98 (0.007)	7%	9%	0.86	0.04
20	150	fixed	0.6	1.000	1.00 (0.000)	13.6 (5.9)	0.99 (0.007)	10%	15%	0.79	0.06
20	150	per task	0.3	0.991	0.96 (0.012)	10.5 (4.6)	0.99 (0.005)	11%	15%	0.79	0.05
60	45	per task	0.6	0.999	0.99 (0.006)	49.0 (8.3)	1.00 (0.000)	29%	41%	0.50	0.05
20	65	per task	0.6	0.997	0.98 (0.007)	18.0 (5.9)	1.00 (0.003)	19%	30%	0.61	0.09

Table 3: Known-truth simulation, 300 boards per scenario. Kept, all: the share of boards on which every truly undominated system is kept in the frontier set. This is the guarantee of Proposition 2. The target is at least 0.95. Monte Carlo standard errors are in brackets after the kept-all and band-covers shares, and are at most 0.013 near 0.95. Costs: whether per-task costs are random or fixed. Corr.: the correlation between a system's cost and its ability in the simulated population. Kept, each: the share of truly undominated systems that are kept, taken one by one. Size: the mean size of the set, with the size of the true frontier in brackets. Band covers: the share of boards on which the band of Section 3.2 contains the true frontier curve at every evaluated cost. Plain frontier: the share of its members that are truly dominated, and the share of the true frontier that it misses. Stability: bootstrap frontier stability. This is the share of task resamples in which a system is on the observed frontier, averaged over the truly undominated systems and averaged over the truly dominated systems.

finds on CORE-Bench hard.

Section S11 of the supplement puts the set beside the alternatives on five of the ten scenarios. The ranges below cover all ten. The joint margin keeps the whole true frontier 0.95 to 1.00 of the time. The exact joint set does the same, at a larger size. The two per-system variants fall to 0.90 on some boards. The two shortcuts fail where the boards are small or crowded, and that is where a leaderboard needs the set most. Unadjusted pairwise tests keep the whole true frontier 0.77 to 0.98 of the time, and the frequency rule keeps it 0.46 to 0.98. Neither one accounts for the number of rivals a system faces. With 20 systems on 50 tasks, the sets hold 16.9 (joint), 18.1 (exact joint), 12.3 (unadjusted) and 10.2 (frequency) systems, against a true frontier of 5.8.

5.1 Stress tests

The base design leaves out features of real boards that the guarantees depend on. Table 4 adds them one at a time to the 20-system board of 150 tasks (200 for the clustered one). Near-copy systems: six of the 20 are degraded copies of another system. Each copy turns a pass of its parent into a failure with probability 0.1, as when one agent is run with a smaller model or a lower effort. In one run the copies cost 3% more than their parents, so a copy can only be dominated. In the other run they cost 3% less, so a copy is a cheap, highly correlated rival that can itself belong on the frontier. Cost-outcome dependence: failed tasks cost twice as much as passed ones, as when an agent uses up many tokens on a task and then fails it. Nearly tied costs: true mean costs are a few per cent apart, closer than the standard error of a mean cost. Clustered tasks: 200 tasks in 40 groups of five. The tasks in a group share a difficulty shift and a system-by-group effect, so tasks are not independent draws. Strongly clustered tasks: 150 tasks in 15 groups of ten, with group and system-by-group effects of about 1.2 on the logit scale. So one system's outcomes within a group are strongly alike. Heavy-tailed costs: per-task costs have a coefficient of variation of 2.5 in place

Scenario	Standard set kept, all	size	Tie-robust set kept, all	size	True frontier	Band covers	Best set true best in	competitor costs no more
Reference: 20 systems, 150 tasks	0.95 (0.012)	13.4	1.00 (0.000)	15.5	5.6	0.98	1.00	0.99
Near-copies	0.98 (0.008)	13.1	1.00 (0.000)	15.8	4.9	0.99	1.00	1.00
Cost-outcome dependence	0.94 (0.014)	12.4	1.00 (0.000)	14.8	5.3	0.98	1.00	0.98
Near-tied costs	0.26 (0.025)	11.1	1.00 (0.000)	20.0	5.7	0.98	1.00	0.87
Clustered tasks	0.95 (0.012)	12.8	1.00 (0.000)	14.9	5.8	0.94	1.00	1.00
groups resampled	0.97 (0.010)	14.0				0.98		
Cheaper near-copies	0.83 (0.022)	13.4	1.00 (0.003)	15.8	6.4	0.97	1.00	0.99
Heavy-tailed costs	0.89 (0.018)	13.3	1.00 (0.000)	18.2	5.6	0.96	1.00	0.97
Strongly clustered tasks	0.93 (0.014)	14.1	0.99 (0.006)	16.1	5.8	0.71	0.97	0.99
groups resampled	0.99 (0.006)	17.9				0.97		
Missing cost logs	0.93 (0.015)	12.5	1.00 (0.000)	14.9	5.4	0.89	1.00	1.00

Table 4: Stress tests on known truth, 300 boards per row, 20 systems. Monte Carlo standard errors are in brackets. Kept, all: the share of boards on which every truly undominated system is kept. Standard set: the set of Section 3.3. Tie-robust set: the set of Proposition 4. Best set: the share of boards on which the true best system is in the best set, and on which the cheapest unresolved competitor of the best costs no more than the true best. Clustered and strongly clustered tasks: the first row resamples tasks, and the second row resamples the groups, for the set and for the band.

of 0.8. Missing cost logs: failed tasks cost twice as much as passed ones, and 30% of the failed tasks have no logged cost. The board prices them at the system's mean over its logged tasks, as the audit does. So every system's mean cost is biased down, by a different amount for each. Five of the eight stresses leave the standard set within about one and a half Monte Carlo standard errors of the nominal 0.95. They are costlier near-copies (0.98), outcome-dependent costs (0.94), clustered tasks (0.95), strongly clustered tasks (0.93) and missing cost logs (0.93). On the reference board the figure is 0.95. Resampling the groups instead of the tasks gives 0.97 on the clustered boards and 0.99 on the strongly clustered ones. The set sizes are 14.0 and 17.9 systems, against 12.8 and 14.1 when tasks are resampled. The set is less exposed to clustering than one might fear. A plausible reason is that group shifts shared by all systems cancel in a paired difference. That is a guess, not a result, and only two clustering strengths were tried. The band is far more exposed, since it uses each system's own accuracy. With tasks resampled, it covers the true curve in 0.94 of the clustered boards and in only 0.71 of the strongly clustered ones. With groups resampled, it covers 0.98 and 0.97, and the strongly clustered band is then 48 points wide instead of 27. Where groups can be named, the band should resample them (`frontier_band(clusters=...)` in honestboard 0.6.1).

Missing cost logs do not visibly hurt the tie-robust set, and they leave the standard set near the nominal level, but they do hurt the band. In this simulation, 30% of the failed tasks are unlogged and priced at the system's mean. The standard set keeps the frontier in 0.93 of boards and the tie-robust set in 1.00. But the band covers the true curve in only 0.89, because its cost intervals are centred on means that are biased down. On HAL, 1.5% of system-task pairs are imputed. That is a small share compared with the failed tasks simulated here. Also, a task that failed before its first call probably costs less than the mean it is priced at, and that would bias the other way. We cannot tell which way the HAL bias runs.

Three stresses push the standard set clearly below 0.95. Two of them are near ties in cost. With nearly tied costs, the set contains the whole true frontier in 0.26 of boards, and a given truly undominated system in only 0.77. With cheaper near-copies, it contains the frontier in 0.83 of boards (a given system in 0.97). A copy that costs 3% less than its parent can belong on the true frontier. But when the sample cost order puts the parent first, the parent becomes the copy's rival, and its paired lead over the copy has a tiny standard error. The third is heavy-tailed costs. The

Benchmark	Tasks	Systems	Mean set size	B's frontier members kept by A's frontier set	B's frontier members kept by A's plain frontier	B's frontier members kept by size-matched random
ColBench backend	1000	22	16.6	1.00	0.82	0.93
ColBench frontend	100	14	7.0	1.00	0.66	0.77
GAIA	165	36	24.2	1.00	0.60	0.83
Online-Mind2Web†	300	32	27.0	1.00	0.71	0.94
ScienceAgentBench	102	25	21.5	1.00	0.63	0.94
USACO	307	12	7.0	1.00	0.99	0.99

Table 5: Stability check on the HAL boards with at least 100 tasks, 200 random half splits each. The frontier set and the plain frontier are computed on half A. The target is the systems on half B's own observed frontier. Columns: the share of those systems kept by A's frontier set, by A's plain frontier, and by a size-matched random baseline. The baseline is A's plain frontier plus randomly chosen systems, up to the size of A's frontier set. The fourth column is the mean size of A's set.

standard set keeps the frontier in 0.89 of boards, which is below the nominal level by about three Monte Carlo standard errors. We have not found the cause for sure. A plausible reason is that the sample cost order is unreliable when mean costs are this skewed, as in the near-tie failures. The tie-robust set does not rely on the sample cost order, and it repairs the failure. We did not check whether more tasks restore the level. This regime is extreme compared with the data. Across the systems with per-task costs on the HAL and HELM boards, the median coefficient of variation of a system's per-task cost is 0.70, the 90th percentile is 1.3, and the largest is 3.5. And 0.5% of the systems reach 2.5.

The tie-robust set keeps the frontier in 1.00, 1.00 and 1.00 of boards in these three. In the nearly tied scenario its size is 20.0 of 20. When true costs are closer than their noise, no system is surely cheaper than another. The data cannot resolve the frontier, and the set says so by returning the whole board. The band does not use the cost order. It still covers the true curve in 0.98 of the nearly tied boards, at a median width of 68 points. The cheapest unresolved competitor of the best costs no more than the true best in 0.87 of them, against 0.97 to 1.00 elsewhere, because it also leans on the cost order. The tie-robust set is conservative in the other scenarios as well. It keeps the whole true frontier in at least 0.99 of boards in every scenario. Across the other scenarios its size is up to 37% above the standard set's (14 to 37% in Table 4), and the largest excess is in the heavy-tailed scenario. We can read from the data whether a real board resembles the tied cases. The share of system pairs whose cost order the paired comparison resolves is 0.01 in the nearly tied scenario and 0.82 in the base design. On the HAL boards with per-task costs, it is 0.74 to 0.98. Most HAL boards look like the base design in this respect. But on CORE-Bench hard the tie-robust set is the whole board (Table 6). This is a caution that the cost order there is only partly resolved.

The crowded board of Section 5, 60 systems on 45 tasks, tests the fixed-K theory harder than any other case, and CORE-Bench hard has that shape. Proposition 2 lets T grow with K fixed. So on such a board the guarantee is supported by simulation and not by the theorem. There the standard set kept the whole true frontier in 0.99 of boards, at a size of 49.0 of 60 systems.

5.2 Held-out halves of the real boards

Simulation has a known truth, but the world in it is made up. Table 5 repeats the frontier check on the real boards with at least 100 tasks. We split the tasks at random into two halves, 200 times. We compute the frontier set and the plain frontier on one half. The target is the systems on the other half's own observed frontier. This is a stability check, not a coverage check. Half B's frontier is itself a noisy estimate, so even a perfect method would not keep all of its members. Also, what a

Benchmark	Systems	Tasks	Observed frontier	Frontier set (share)	Tie-robust set	Best set	Top system (%, $)	Cheapest unresolved (%, $)
assistantbench	32	33	3	32 (100%)	32	32	15.2, 4.39	3.0, 0.02
colbench backend programming	22	1000	9	14 (64%)	15	5	43.7, 0.20	38.6, 0.02
colbench frontend design	14	100	4	7 (50%)	7	5	82.9, 0.12	81.6, 0.03
corebench hard	60	45	8	53 (88%)	60	1	75.6, 6.08	75.6, 6.08
gaia	36	165	10	22 (61%)	24	7	74.5, 1.13	64.2, 0.74
online mind2web†	32	300	6	23 (72%)	23	16	42.3, 0.57	36.3, 0.30
scicode	37	65	3	37 (100%)	37	37	9.2, 0.72	1.5, 0.00
scienceagentbench	25	102	4	21 (84%)	21	17	33.3, 0.06	27.5, 0.02
swebench verified mini	39	50	6	26 (67%)	32	11	72.0, 1.57	54.0, 0.67
taubench airline	47	50	7	40 (85%)	43	26	66.0, 3.14	44.0, 0.04
usaco	12	307	4	5 (42%)	5	1	69.7, 0.11	69.7, 0.11

Table 6: Frontier audit of HAL's benchmarks at $\alpha = 0.05$, $B = 4000$ task resamples. Observed frontier: systems that no cheaper system beats on the point estimates. Frontier set: systems that cannot be excluded from the frontier. Share: that count over the number of systems on the board. Tie-robust set: the set of Proposition 4. Best set: systems that cannot be excluded as the most accurate. Top system: the most accurate system. Cheapest unresolved: the cheapest unresolved competitor of the best. We give both of these with their accuracy and their cost per task in dollars. Costs are dollars per task, and we repriced them from the call logs with HAL's price table as read on 13 October 2026. A frontier under one price table is a statement about those prices. It is not advice on what to buy. †: no per-task costs exist on this board. Each run's repriced total is spread evenly over its tasks and treated as fixed. So there is no cost uncertainty, and the tie-robust set is defined to be the frontier set.

large set keeps is partly a matter of its size. So we read the retention beside a size-matched random baseline. A's frontier set keeps 1.00 of B's frontier members on every board, A's plain frontier keeps 0.60 to 0.99, and the size-matched random baseline keeps 0.77 to 0.99. So the set beats random padding of the same size on every board, by 0.01 to 0.23. The sets hold 50 to 86% of the systems, depending on the board. The set is not merely large: it concentrates on the systems that tend to be on a frontier next time. The two halves share the task pool, so the check covers task-sampling noise only. This is the noise the method models. The winner's-curse part of the same split is in supplement S3.

6 Audit of the Holistic Agent Leaderboard

HAL's own frontiers change under consistent costing. Before any uncertainty enters, the point-estimate frontier itself depends on how costs are counted and priced. Here "HAL's published frontier" means the frontier of the kept runs under HAL's reported totals. The runs we dropped, such as those of Claude Code, are absent from it, and they may sit on HAL's own frontier. Table 7 draws it three ways. Going from HAL's reported totals to once-counted, repriced costs changes the frontier on 8 of 11 boards. 10 systems leave it and 18 join. That is 28 changes against 56 systems on the reported frontiers (50%). Removing the duplicate spans alone moves 4 systems on 2 boards. The reason is that the doubling is shared by the agents that use the same software stack, so it mostly leaves their order intact. Pricing every run under one table moves 24 more. There are two reasons. HAL's totals carry the prices of the month each run was made. Also, 5 runs are reported at a total near zero although their call logs price at more than a cent per task, with all of them sitting on HAL's frontier for that reason alone. So a reader of HAL's pages is looking at frontiers that mix two counting rules and a year of price changes. The supplement draws the two boards that change most.

Table 6 and Figure 1 give the audit itself. The supplement also draws the set sizes. Six readings and two sensitivity checks follow.

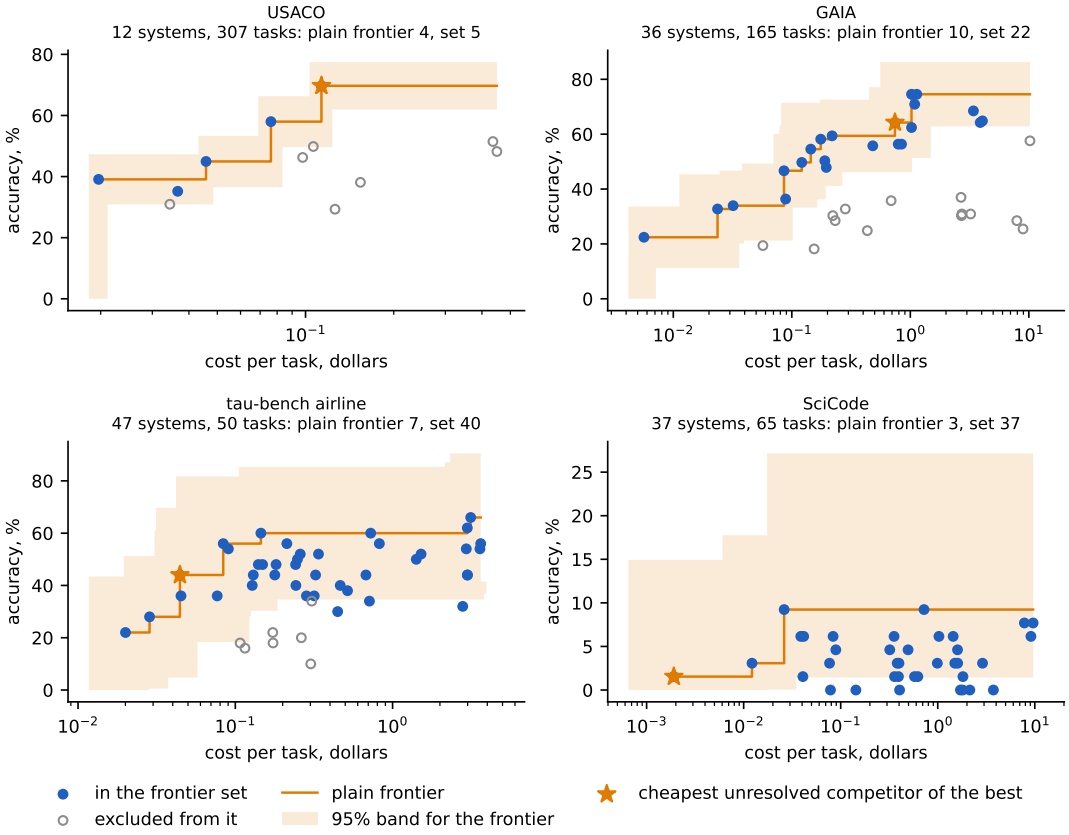


Figure 1: Cost per task against accuracy on four of the eleven HAL benchmarks. On USACO the set is close to the plain frontier. On GAIA and tau-bench airline it is much wider. On SciCode nothing can be excluded. The orange staircase is the observed frontier. Filled points are systems that cannot be excluded from the frontier at the 5% level. Hollow points can be excluded. The star is the cheapest unresolved competitor of the best: the cheapest system that cannot be excluded as the most accurate. The shaded region is the 95% simultaneous band for the frontier curve (Section 3.2). Costs of zero are drawn at 10^{-4} dollars. All eleven boards are in the supplement.

Benchmark	Plain frontier size under reported	once-counted	ours	Reported to ours lost	gained	Changes due to counting	pricing
assistantbench	3	3	3	1	1	0	2
colbench backend programming	9	9	9	0	0	0	0
colbench frontend design	3	3	4	0	1	0	1
corebench hard	5	5	8	2	5	0	7
gaia	5	5	10	1	6	0	7
online mind2web[†]	6	6	6	0	0	0	0
scicode	3	3	3	1	1	0	2
scienceagentbench	8	6	4	4	0	2	2
swebench verified mini	4	6	6	0	2	2	0
taubench airline	6	6	7	1	2	0	3
usaco	4	4	4	0	0	0	0
All	56	56	64	10	18	4	24

Table 7: The plain frontier of each board under three costings of the same kept runs. The first is HAL's reported totals. The second is those totals with the duplicate spans removed (HAL's historical prices kept). The third is the once-counted, repriced costs of the audit. Lost and gained: systems that leave or join the frontier going from HAL's costing to ours. The last two columns split the changes into those made by counting each call once and those made by repricing after that. A frontier that changes under a costing rule is a finding about the rule. It is not advice on what to buy.

The frontier is far less certain than the picture suggests. Across the 11 benchmarks, 64 of 356 systems sit on the observed frontier, and 280 cannot be excluded from it. A pooled share hides the spread, so Table 6 gives the share board by board. It runs from 42% to 100%. Without the 2 boards on which nothing can be excluded, the pooled share is 74%. The simultaneous margin is 3.04 to 3.77 standard errors of a paired difference. On a board of 45 to 50 tasks, a paired standard error is several accuracy points. The smallest lead of an excluded system is 26 to 28 points on those boards, and 5 to 10 points on the boards with 300 or more tasks.

The band says the same thing as a curve. The shaded region in Figure 1 is the 95% band of Section 3.2. Its half-width at a frontier member is 2.90 to 3.67 of that member's own standard error. Its width, taken as the median over the costs at which it is evaluated, runs from 6 points on ColBench frontend to 62 points on CORE-Bench hard. A reader who wants one picture of what a board can and cannot say about the best accuracy available at each price will find it here. The picture is less confident than the staircase. Take the best-measured boards: ColBench backend with 1000 tasks, and USACO and Online-Mind2Web with about 300. Even their bands are 10 to 18 points. The ColBench frontend band is narrow, at 6 points, only because its continuous scores vary little from task to task. On the boards with 33 to 65 tasks, the band is 26 to 62 points wide. That is a large part of the whole range of accuracies. A step of the staircase can be trusted only where it is taller than the band.

Task count decides what can be said. The set is closest to the observed frontier, measured as set size over frontier size, on the two boards USACO (307 tasks: frontier 4, set 5) and ColBench backend programming (1000 tasks: frontier 9, set 14). The two boards where nothing can be excluded are AssistantBench (33 tasks, accuracies up to 15.2%) and SciCode (65 tasks, up to 9.2%). With success rates that low and samples that small, no cheaper system can be shown to

			Systems kept					Tasks needed	
Benchmark	Tasks	Plain frontier	joint set	largest-margin	per-system	exact, per system	exact, joint	to exclude half	for a 5-point lead
assistantbench	33	3	32	32	32	32	32	171	442
colbench backend programming	1000	9	14	12	12	12	15	2,857	1,012
colbench frontend design	100	4	7	7	7	7	7	7,863	52
corebench hard	45	8	53	49	49	52	57	408	1,630
gaia	165	10	22	20	20	21	23	606	1,699
online mind2web†	300	6	23	21	19	20	24	1,811	1,370
scicode	65	3	37	37	37	37	37	265	365
scienceagentbench	102	4	21	18	18	20	21	529	1,169
swebench verified mini	50	6	26	23	17	24	27	122	1,179
taubench airline	50	7	40	35	32	39	42	291	1,748
usaco	307	4	5	5	5	5	6	1,057	697

Table 8: Five frontier sets and two resolution numbers per board. Joint set: the confidence set of Section 3.3. Largest-margin and per-system: its two per-system variants. Exact, per system and exact, joint: the finite-sample sets of Section 3.3. Tasks to exclude half: the task count at which half of the systems now in the set but off the plain frontier would leave it, if their gaps to their strongest cheaper rival stayed as observed. Tasks for a 5-point lead: the task count at which a system trailing a cheaper rival by five accuracy points would be excluded, given the board's median paired noise.

beat any other. CORE-Bench hard, with 60 systems on 45 tasks, keeps 53 of them in the set.

How far the boards are from resolving their frontiers. Table 8 turns the set sizes into task counts. Excluding half of the systems that now sit in the set but off the plain frontier would take 122 to 7,863 tasks across the boards that have such systems. Excluding a system that trails a cheaper rival by five points would take 52 to 1,748. Compared like for like, the exact sets are the larger ones. The exact joint set holds 291 systems against the confidence set's 280. It is larger on 7 boards and smaller on 0 boards. The exact per-system set holds 269 against the per-system set's 248. This is the usual trade between a finite-sample guarantee and power. The two per-system variants of the confidence set hold 259 and 248 systems against its 280. So the joint guarantee costs a few dozen systems across the audit. The two boards with the largest "tasks to exclude half" are ColBench backend and frontend. On these, the few systems not excluded are near-ties with their rivals, and no task count resolves such near-ties quickly.

The best is often clear even when the frontier is not. The best set is a single system on 2 boards (CORE-Bench hard and USACO). On those boards the frontier set is larger. For the best set, a system's rivals are all the other systems. For the frontier set, its rivals are only the cheaper ones. So the frontier set is at least as large as the best set on every HAL board. It is equal only on AssistantBench and SciCode, where both are the whole board. A cheap, weak system stays in it whenever no cheaper system beats it by the margin. The nesting is an observation about these boards, not a guarantee, because the two margins are different quantiles. The frontier set is a statement about every system's own neighbourhood of cheaper rivals. On most boards, those neighbourhoods are measured with little data.

The cheapest unresolved competitor of the best is usually not the top system. On 7 of 11 boards, the cheapest system that cannot be ruled out as the best is a different and cheaper system than the most accurate one. On 2 further boards (AssistantBench and SciCode) the best set is the whole board. There the cheapest unresolved competitor is simply the cheapest system, and the statement is empty. The gaps are large. On tau-bench airline the top system costs $3.14 per task and scores 66.0%, while the cheapest unresolved competitor costs $0.04 and scores 44.0%. On GAIA the pair is $1.13 at 74.5% against $0.74 at 64.2%. On SWE-bench Verified mini it is

$1.57 at 72.0% against $0.67 at 54.0%. This does not say the cheap system is as good. It says that 50 tasks cannot show that it is worse. That holds at a level that accounts for all 47 systems being compared at once. More tasks are what would settle it.

How much the remaining costing rules matter. Two more costing choices could have driven the results. One is pricing cache reads and writes at the table's cache price. The other is setting the cost of a task with no logged model call to the run's mean cost. The supplement recomputes the boards from the stored token counts under the opposite choice for each. These are the full input price for cached tokens, and a cost of zero for an unlogged task. Pricing cached tokens at the full input price raises the mean cost per task by up to a factor of 2.0, on SWE-bench Verified mini. It moves 8 systems on or off the plain frontier on 3 boards, and 7 in or out of the frontier set. Costing an unlogged task at zero moves 3 systems on the plain frontier and 2 in the set. So the cache-price rule moves more systems on the plain frontier than the double count does by itself (4). It moves far fewer than the 24 moved by pricing at the prices of each run's own date. The set sizes, which are the audit's main result, change by a few systems. Over the 10 boards with per-task costs, the sets hold 257 of 324 systems under the audit's rules. They hold 260 with cached tokens at the full input price, and 257 with unlogged tasks at zero. No single board changes by more than 2 systems. Online-Mind2Web, which has run totals only, is not in these counts.

If the standard errors are too small. Every set above treats tasks as independent draws and one run as a system's average. Two things can make its standard errors too small. One is tasks that come in groups. The other is variation between runs that the task-level noise does not contain (Section 8). Both act on the studentised leads as one common factor f on every standard error. So the set at a given f can be read off the stored leads without a new bootstrap. A system stays unless its strongest cheaper rival's lead, divided by f, exceeds the board's margin. A design effect of d is a factor $f = \sqrt{d}$, and the effective number of tasks is T/f^2. Table 9 gives the set size at $f = 1$, 1.25, 1.5 and 2 for every board, and the smallest f at which no system is excluded. On HAL the sets hold 280, 297, 316 and 330 of 356 systems at $f = 1$, 1.25, 1.5 and 2. On HELM they hold 51, 58, 62 and 75 of 112. Two kinds of departure act differently. A design effect from clustered tasks is multiplicative, and it is what f stands for. A run-level standard deviation is additive: it adds the same variance to every comparison, so it matters most on the boards that are best measured. The last column of Table 9 gives the HAL sets under the run-level excess measured from the rerun pairs (Section 8). Board by board it is larger than $f = 2$ on the best-measured boards, and smaller on the boards with 45 to 50 tasks. The pooled total of 324 falls between $f = 1.5$ and $f = 2$ only because the two effects offset. At $f = 2$ the set is the whole board on 5 of the 11 HAL boards. These are AssistantBench, SciCode, CORE-Bench hard, tau-bench airline and ScienceAgentBench, which have 33 to 102 tasks. On none of the HELM boards is the set the whole board at that point. Those boards need f of 3.0 to 7.5 before nothing can be excluded. Boards with many tasks or widely spread systems lose the least, but they do lose: at $f = 2$ the sets still leave some systems out (Table 9). These are the standard sets. The tie-robust set is a step-down procedure, so it cannot be read off the stored leads, and we do not inflate it here.

Some boards have task groups that we can name. For these, the cluster bootstrap gives an observed factor. It resamples groups and keeps the task-level standard errors. The factor is the ratio of its margin to the task margin. ScienceAgentBench's 102 tasks come from 30 source repositories and four science domains. Resampling repositories moves the margin from 3.46 to 4.11 standard errors, a factor of about 1.2, and the set from 21 to 21 systems. Resampling domains gives a margin of 4.56, a factor of 1.3, and a set of 22. With repositories resampled, the band for ScienceAgentBench

Board	Systems	Tasks	Set size at $f =$ 1	1.25	1.5	2	Whole board at $f =$	Measured run-level sd
AssistantBench	32	33	32	32	32	32	1.0	32
ColBench backend	22	1000	14	15	18	19	3.6	22
ColBench frontend	14	100	7	7	7	7	5.4	13
CORE-Bench hard	60	45	53	57	59	60	1.5	58
GAIA	36	165	22	24	25	28	3.4	26
Online-Mind2Web†	32	300	23	27	29	31	2.1	32
SciCode	37	65	37	37	37	37	1.0	37
ScienceAgentBench	25	102	21	21	23	25	1.9	24
SWE-bench Verified mini	39	50	26	28	32	36	3.1	29
tau-bench airline	47	50	40	42	46	47	1.8	42
USACO	12	307	5	7	8	8	4.6	9
GPQA	28	446	17	18	19	24	3.0	–
IFEval	28	541	11	15	17	24	3.9	–
MMLU-Pro	28	1000	13	15	16	17	6.2	–
Omni-MATH	28	1000	10	10	10	10	7.5	–

Table 9: Frontier set size if every standard error were f times the one used. The HAL boards are above the line and the HELM boards below it. $f = 1$ is the set of Tables 6 and 10. Whole board: the smallest f at which every system stays in the set. Measured run-level sd: the HAL set under the run-level excess measured from the rerun pairs (Section 8). †: cost is a fixed number per system. The step from $f = 1$ to $f = 1.5$ is the effect of a design effect of 2.25, or of a run-level standard deviation about as large as the standard error already used.

is 43 points wide, against 34 with tasks resampled. SWE-bench Verified mini and USACO have only two and four groups. Their factors are 0.9 and 1.8. These say more about having few groups than about dependence. Online-Mind2Web's websites are named in a gated dataset that we could not read, so that board has no cluster check.

7 A second leaderboard: HELM Capabilities

HAL is one leaderboard. It is built from agent runs, most of its boards have small task counts, and we rebuilt the costs ourselves. To see whether the findings and the behaviour of the method carry over, we repeat the audit on a leaderboard of a different kind. HELM Capabilities [17, 29] scores language models on five fixed benchmarks. It uses one request per task and no agent scaffold. For every model and instance, it publishes the outcome and the number of prompt and completion tokens. Its 68 models arrive release by release, in sixteen releases from March to November 2025. So the releases give an arrival order for the certificates.

Cost. HELM publishes no cost, so we price the recorded tokens of each task at list prices. The rule is fixed in advance. OpenAI and Anthropic models get their first-party price (the HAL price table). Amazon Nova gets its Bedrock price. Open-weight models get the Together AI list price, because HELM ran them through Together. A model with no such price is left out, and we do not price it at some other host (supplement S13). The cost is only as good as the token counts. So a model enters a board only when its recorded counts can serve as the counts we bill. WildBench is left out because it records no prompt tokens. Of the 68 models, 12 record no completion tokens (the Google and DeepSeek runs and some Writer runs), 11 hide their reasoning tokens (the OpenAI o-series and GPT-5, Claude with extended thinking, Grok 4, Grok 3 mini and Qwen3-Next thinking),

Benchmark	Systems	Tasks	Observed frontier	Frontier set (share)	Tie-robust set	Best set	Top system (%, $)	Cheapest unresolved (%, $)
GPQA	28	446	6	17 (61%)	18	6	68.6, 0.0090	68.4, 0.0008
IFEval	28	541	5	11 (39%)	14	2	91.8, 0.0324	90.4, 0.0005
MMLU-Pro	28	1000	6	13 (46%)	13	2	86.9, 0.0066	86.9, 0.0066
Omni-MATH	28	1000	6	10 (36%)	10	2	68.8, 0.0025	68.8, 0.0025

Table 10: Frontier audit of the four HELM Capabilities boards at $\alpha = 0.05$ with $B = 4000$ task resamples. The columns are as in Table 6. Cost is dollars per task. It comes from the recorded prompt and completion tokens at list prices read on 14 October 2026 (first-party prices for OpenAI and Anthropic, Together AI for open-weight models), under the rule in the text. As for HAL, a frontier here is a statement about these prices and this host. It is not advice on what to buy.

and 17 have no price under the rule. That leaves 28 models on each of four boards: GPQA, IFEval, MMLU-Pro and Omni-MATH. They have 446 to 1000 tasks. So these boards are not the cost-accuracy frontier of language models. Reasoning models whose reasoning is hidden are absent. Open models that print their reasoning, such as gpt-oss and GLM-4.5-Air, are in. The host also matters. When we reprice the 9 open-weight models that DeepInfra also lists, their cost falls to about 0.46 of the Together figure. This moves 1 to 5 systems on or off each board's plain frontier, and 2 to 5 in or out of its frontier set (supplement S13). A frontier built on list prices is a statement about those prices. The boards are a second test of the method, with a different shape. They are not a claim about which model to buy.

The frontier is better resolved than on HAL. Of 112 system entries, 23 sit on observed frontiers. 51 (46%) cannot be excluded from them. By board, this share is 36% to 61%, against 42% to 100% on HAL (Table 10). The boards have 446 to 1000 tasks, and the models differ a lot in cost and accuracy. The margin is 3.50 to 3.55 standard errors, inside the range of HAL's (3.04 to 3.77). But the standard errors themselves are small, so the band is 7 to 15 points wide. The tie-robust set holds 55 systems, against 51 for the standard set. What is left unresolved is a few close competitors. To exclude half of them would take 1,465 to 6,305 tasks, against 122 to 7,863 on HAL. On 2 of the 4 boards, the cheapest unresolved competitor of the best is a cheaper model than the top one. On IFEval the top model, Claude Opus 4, scores 91.8% at $0.0324 per task. The cheapest model that cannot be ruled out as the best, GPT-4.1 mini, scores 90.4% at $0.0005. On MMLU-Pro and Omni-MATH the top model is itself the cheapest unresolved competitor. The sets are narrower because the boards are larger and the models are further apart in cost and accuracy. This is the paper's message about task counts, seen from the other side.

Certificates in release order. We take the models of each board in release order. Across the four boards, 28 of the 112 model entries are testable entrants, and another 25 are the cheapest so far. Of the entrants, 17 are certified as record claims under alpha-spending over each board's chain. 21 pass the 5% level once the per-date factor of Appendix A is applied. 17 are certified as Pareto members in the normal form. Models released together share a date, so the per-date factor applies.

7.1 Do certified entrants keep their lead on fresh tasks?

A certificate is a statement about the tasks it was computed on. To test it on tasks the procedure never saw, we split the tasks of every board with at least 100 tasks (6 HAL boards and 4 HELM boards) at random into two halves. We do this 40 times. We run the arrival chain of Appendix A on one half alone, with each system's cost fixed at its mean on that half. We test every frontier entrant against the most accurate cheaper system that came before it. Then we measure the entrant's lead

Group of entrants	cases	pairs	lead on D	lead on V	positive on V	pairs with positive mean lead on V (exact 95% interval)
HAL						
Certified (record claim)	96	12	13.6	10.0	0.97	10 of 12 [0.52, 0.98]
Nominal only	110	24	9.0	7.3	0.94	18 of 24 [0.53, 0.90]
Neither	1110	145	4.6	4.6	0.80	108 of 145 [0.67, 0.81]
HELM Capabilities						
Certified (record claim)	506	39	14.2	14.4	1.00	38 of 39 [0.87, 1.00]
Nominal only	177	29	6.6	6.6	0.98	27 of 29 [0.77, 0.99]
Neither	426	35	3.5	4.4	0.85	30 of 35 [0.70, 0.95]

Table 11: Fresh-task replication of the entrant certificates. There are 40 random half splits per board, on the boards with at least 100 tasks, and cases are pooled over splits and boards. Certified: the record claim is certified under alpha-spending over the chain on the discovery half D. Nominal only: the per-date corrected p-value is at most 5% but the entrant is not certified. Neither: all the rest. Lead on D and on V: the mean accuracy lead over the holder, in points, on the discovery half and on the fresh half V. Cases: the entrant-holder pairs tested, counted once per split. Pairs: how many distinct pairs these cases are. Positive on V: the share of cases whose lead on V is positive. Last column: the number of pairs whose mean lead on V, over the splits in which the pair is in the group, is positive, with an exact 95% interval that treats pairs as independent. This is the figure to read. Board-by-board and pair-by-pair results, with the number of splits per pair, are in supplement S14.

over that holder on the other half. An entrant and the holder it is tested against make a pair. The same pair can be certified in several splits, and many pairs are certified in only one or two of the splits.

Counted case by case, certified entrants keep a positive lead on the fresh half in 0.97 of cases on HAL and 1.00 on HELM. The lead is significantly positive in 0.83 and 0.96 (Table 11). The cases are not independent. A pair counts as a certified case in every split in which it is certified on the discovery half. The 96 certified HAL cases are 12 distinct pairs on 3 boards, and the 506 HELM cases are 39 pairs on 4 boards. These pairs are not the entrants certified on the full boards (Section 3.5). They are the pairs certified on a random half in at least one split. Most are certified in few splits. On HAL the median pair is certified in 2 of the 40 splits, and 7 of the 12 pairs are certified in two splits or fewer. On HELM the median is 11 splits, and 11 of the 39 pairs are certified in two splits or fewer. A pair certified in one split is one coin flip, not a replication. So the last column of the table counts pairs, not cases. For each pair we take its mean lead on the fresh half over the splits in which it is certified, and we count the pairs for which this mean is positive. For certified entrants this holds for 10 of 12 pairs on HAL (exact 95% interval 0.52 to 0.98) and for 38 of 39 on HELM (0.87 to 1.00). For entrants that are neither certified nor nominally significant it holds for 108 of 145 pairs on HAL (0.67 to 0.81) and 30 of 35 on HELM (0.70 to 0.95). On both leaderboards the intervals for certified pairs and for the rest overlap. A rank test on the share of splits with a positive lead separates the two groups on HELM (one-sided $p = 0.004$) but not on HAL ($p = 0.091$). The intervals and the test treat pairs as independent, although pairs share boards, holders and tasks. So they show precision, and they do not guarantee coverage. Two further cautions apply. First, the groups are defined by evidence on the discovery half, so they differ in lead on that half. The lead is 13.6 points for certified entrants on HAL, against 4.6 for the rest.

A larger lead replicates more often, no matter why the entrant was certified. Among entrants with a lead of at least five points on the discovery half, certified entrants keep a positive lead in 0.97 of cases on HAL and 1.00 on HELM. For entrants that are neither certified nor nominally significant, the figures are 0.82 and 0.98. In the same group of entrants, certified entrants keep a significantly positive lead in 0.83 and 0.96 of cases, against 0.23 and 0.75. Second, a positive lead on fresh tasks is a low bar. 0.80 of the entrants that are neither certified nor nominally significant keep one on HAL, and 0.85 on HELM. A certificate is meant to protect against a lead that is only noise. On HELM the pair-level evidence is consistent with this, and on HAL the data cannot separate the groups. The winner's curse remains. On HAL the lead of certified entrants shrinks from 13.6 points on the discovery half to 10.0 on the fresh half, as in Appendix B. On HELM it does not shrink (14.2 to 14.4). There the gaps are large compared with the noise. The halves share the task pool and the systems. So this tests the record claim against the holder on new tasks from the same pool. It does not test Pareto membership, and it does not test tasks from a different benchmark.

8 Limitations

Run-to-run noise. Each system has one run. The standard error of a single run comes from the sample variance of its task outcomes. So it already contains the independent run-to-run noise of each task. What is missing is any excess beyond that. The excess appears when a run is better or worse than its configuration's average for reasons shared across its tasks. HAL holds 8 pairs of runs of one configuration (same agent, model and settings) on 3 boards. These pairs measure the excess directly (supplement S10). The two runs of a pair differ by 6.4 accuracy points on average and 14.0 at most. In cost they differ by a median factor of 1.09 (at most 1.72). Suppose every task's outcome flipped independently between the two runs, at the observed rate of disagreement. Then the root-mean-square difference would be 5.6 points. The observed value is 8.3. This gives an excess standard deviation of 4.4 points per run. With 8 pairs this is weak evidence (chi-square 13.5 on 8 degrees of freedom, $p = 0.096$), but it is not nothing. Also, 3 of the pairs share a run with another pair, which leaves at most seven degrees of freedom and $p = 0.061$. We carry that excess to every board through the per-task disagreement rate, and we add it to both systems of every paired comparison. An excess of that size would then grow the frontier sets from 280 to 324 systems in all. If the excess is real, the audit's sets are lower bounds. The HAL results are conditional on negligible variation shared across a run's tasks. Table 9 shows the sets for a range of such inflation. The remedy is repeated runs, which the method accommodates by averaging them. We could not add any, because that means running the agents again.

Task clusters. We treat tasks as exchangeable draws. Where tasks cluster (several from one repository, one website, one problem family), the standard errors are optimistic, as in any task-level analysis. In simulation, with 40 groups of five tasks that share a difficulty shift and a system-by-group effect, the set built by resampling tasks contains the whole true frontier in 0.95 of boards. The set built by resampling groups does so in 0.97 (Table 4). The module can resample task groups, but most of HAL's boards offer little to resample. SWE-bench Verified mini's 50 tasks come from 2 repositories, and USACO's 307 come from 4 difficulty divisions. So margins of 3.36 and 5.52 standard errors, against 3.66 and 3.04, say more about having two or four units than about dependence. ScienceAgentBench is the one board with many groups that we could name. Its tasks come from 30 source repositories, and resampling them leaves the set at 21 systems, against 21 (Section 6 and supplement S15). Online-Mind2Web (tasks within websites) probably has many small groups. But the website labels are in a gated dataset, so we ran no cluster bootstrap

there. CORE-Bench hard, GAIA and tau-bench airline have no groups that we could name. In the strongly clustered simulation the band covered the true curve in only 0.71 of boards when tasks were resampled, and in 0.97 when groups were resampled. So the empirical conclusions are conditional on task-level exchangeability, as the tables of Section 6 show for a range of departures from it.

The rest. Costs depend on prices, and prices change. We fix one price table and say which one. A reader who prices a run under another table should rebuild the frontier and not reuse ours. The recount rests on HAL's published call logs, and we reran no agent. HAL's team has confirmed the double counting [19]. It has not confirmed our recount as a whole. A request to do so is drafted in the repository. The HELM costs are recorded tokens at list prices, so they approximate a bill. They exclude the models whose reasoning tokens are not recorded. They also depend on the host chosen for the open-weight prices (Section 7). The certificates and the winner's-curse corrections fix each system's cost at its mean. So they do not correct for selection through noisy costs. Systems may have been tuned on the public tasks. This inflates every accuracy on the board, and it is outside the model. We take outcomes as HAL recorded them. HAL has since disclosed that on tau-bench airline, the episodes of its Generalist agent that never terminated were recorded as failures. So those rows are lower bounds on that scaffold's accuracy [19]. The guarantees of the band and the set are asymptotic. The band needs no condition on the costs. The set is proved for fixed costs, and for per-task costs whose true means are distinct. The tie-robust set removes that condition, at the price of size. On Online-Mind2Web the cost is a run total, treated as fixed. The set uses one margin for the whole board. This is conservative when the systems are many. Per-system margins would give a smaller set, with a weaker guarantee. Finally, the set is a statement about K specific systems on T specific tasks. A new system or a new task changes it.

9 Recommendations

1. Report the frontier set, or the band, beside the frontier. A board that draws a frontier from point estimates should say which systems cannot be excluded from it, at a stated level. It should also say how many tasks that statement rests on, and how many it would take to resolve it (Table 8: from about a hundred to several thousand tasks on the present boards).

2. Publish per-task costs and per-call token counts. Count each model call once. Price every run under one named table, with its date. The confidence set needs per-task costs to carry the cost uncertainty. The audit shows that run totals alone can hide a difference of several times in price between two dates.

3. Where agents built on one model have nearly tied costs, use the tie-robust set. It stays valid when costs tie, and it shows the tie by growing.

4. Report the cheapest unresolved competitor of the best, with its level, beside the top system. Read it as a reason to run more tasks. Do not read it as a purchasing recommendation.

5. Certify frontier entrants with an error budget that is spent over the calendar. Say which claim is certified: the record claim against the most accurate cheaper system (exact), or Pareto membership (the composite test, under its conditional null). Fix costs on arrival, because the guarantee needs it (Appendix A).

10 Conclusion

A cost-accuracy frontier is an estimate. If we treat it as one, with a confidence set in place of a line, HAL's benchmarks say less than the lines suggest, and more than nothing. On some boards the frontier is sharp. On most it is wide. On a few it says nothing until more tasks are run. The tools here let a maintainer say which case a board is in. They hold for tasks drawn from the benchmark's task process, and they are conditional on the systems submitted. They say nothing about tuning on public tasks or about new kinds of tasks. The audit also takes tasks to be independent and one run to be a system's average. If the true standard errors are 1.5 times ours, the HAL standard sets grow from 280 to 316 of 356 systems (Table 9). If the extra variation behind this is real, the sets reported here are lower bounds.

Reproducibility

The paper's public repository (`https://github.com/jopoku16/frontier-leaderboards`) holds the code, the per-system result summaries, the simulations and the audit. The methods are in the open-source package `honestboard` (module `frontier`, version 0.6.1). We do not redistribute the compact per-task records derived from HAL's trace archives, or the raw archives, because the dataset states no licence. The scripts rebuild them from the public archives. The supplement lists every archive with its status. Two scripts of the repository build the HELM boards from HELM's public per-instance results. The price table is in the supplement.

A Certifying a frontier entrant

This appendix gives the certificates for a new frontier entrant and the proofs that go with them. HAL's own timeline is in supplement S2.

A.1 Ingredients from the companion paper

We use five pieces of Banahene and Opoku [5] below. Each is stated here in the form we need.

The selective exact p-value. For a holder a and a challenger b on the same T tasks with pass/fail outcomes, let n_b be the number of tasks b passes and a fails, n_a the reverse, and $n = n_a + n_b$. A leaderboard tests b only because $\bar{y}_b > \bar{y}_a$. That is, $n_b > n_a$. The selective exact p-value is the tail of the sign test given that event, $p_{\text{sel}} = \mathbb{P}(X \geq n_b \mid X > n/2)$ with $X \sim \text{Bin}(n, 1/2)$. For outcomes that are rates in $[0, 1]$, we apply the same conditioning to a sign-flip test of the paired differences. We run that test by Monte Carlo over 200,000 sign patterns.

The conditional task-wise dominance null. This is the null under which p_{sel} is valid. On every task, given the two systems' outcomes on the other tasks, b is no more likely to pass than a. Tasks are independent. It is the natural "b is not better than a" for paired data. Proposition 1 of the companion paper shows that under this null p_{sel} is super-uniform given the record event. This holds as long as the holder was chosen by an event that is increasing in the holder's own outcomes and does not involve b.

The per-date factor and the best-first order. Submissions that share a date are ordered best first. So the challenger tested on a date is the best of that date. This is a second selection, and the record event does not cover it. We multiply the p-value by g, the number of submissions on the challenger's date. This is a Bonferroni factor over the date's members, and it restores the guarantee.

Alpha-spending. The t-th tested step is tested at level $\alpha_t = 6\alpha/(\pi^2 t^2)$. The levels sum to α. So by the union bound, the probability of certifying any null step over the whole chain is at most α. This holds for any number of steps and any dependence. It needs two things. Each step's p-value must be valid given that the step occurred. And the step's level must be settled before its outcomes are looked at.

The online fallback. The t-th step is tested at α_t plus the level of the previous step, when that step was certified [30, 31]. So a run of certified steps builds up budget. The family-wise guarantee is the same. The proof is a disjointness argument over the runs of certified steps that feed each null step.

A.2 The certificates

A live board changes when submissions arrive. A new system b lands on the frontier when it beats every system cheaper than it. The maintainer who then tests it has selected it for exactly that reason, and a test that ignores the selection is biased. The frontier event looks harder than the record event, because it is an intersection of events against several systems. As an event, it is not harder.

Let $C(b) = \{j \text{ earlier} : c_j \le c_b\}$ be the cheaper-or-equal systems when b arrives. Let j^* be the most accurate member of $C(b)$, with ties broken by a rule that does not look at b. Then

$$b \text{ is on the observed frontier on arrival} \quad \Longleftrightarrow \quad \bar{y}_b > \bar{y}_{j^*},$$

because beating the most accurate cheaper system is the same as beating every cheaper system. A tie with a strictly cheaper system leaves b dominated, and a tie at equal cost leaves both on the frontier. Also, b counts as an entrant only when the inequality is strict. So the frontier event is the record event of the companion paper for the single pair (holder j^*, challenger b). The holder was chosen by an event, "j^* is the most accurate system in $C(b)$". This event is increasing in j^*'s own outcomes task by task. Otherwise it depends only on the other cheaper systems, and it does not involve b. These are the conditions of Proposition 1 of that paper.

Proposition 5 (frontier entrant)**.** *Fix the costs before the outcomes are seen. Under the conditional task-wise dominance null for the pair (j^*, b) and independence across tasks, the selective exact p-value $p_{\text{sel}}(b \text{ vs } j^*)$ is super-uniform conditional on the event that b is on the observed frontier on arrival. A rejection at level α certifies that b beats the system that was the most accurate cheaper system when b arrived.*

Remark 3 (what the certificate says, and what it does not)**.** *The null tested is pairwise. Suppose some cheaper system k truly dominates b, while the observed best cheaper system j^* is a weak relative of b. Such a relative is the same agent with a smaller model, a lower effort setting, or a rerun. Its outcomes are correlated with b's. Then b can beat j^* with the stated guarantee and still be truly dominated by k. The test is not of level α for the composite null "some cheaper system dominates b". In simulation, the apparent rival is such a near-copy, and a truly dominating independent system is in the pool. Then a dominated entrant is certified in 3.3 to 34.8% of the cases in which it lands on the frontier. When every cheaper system is independent of it, the rate is 1.1 to 3.6% (20000 boards per case). So the certificate is a record claim restricted to the cheaper pool. It becomes a certificate of Pareto membership only under an added assumption. The assumption is this: whenever b is truly dominated by a cheaper system, it is also task-wise dominated by the observed best cheaper system. One could test b against each cheaper system with the pairwise selective test and require every test to reject. This does not repair the problem, because each pairwise test conditions on its own event and not on the whole frontier event. Conditioning every test on the whole event does repair it, at the price of a normal approximation. That is the next proposition.*

Proposition 6 (Pareto certificate)**.** *Let the vector of accuracies be approximately normal, $\bar{y} \sim N(\theta, \Sigma)$ with Σ the task-level covariance divided by T, and let $E = \{\bar{y}_b \ge \bar{y}_k \text{ for every } k \in C(b)\}$ be the frontier event. For each $k \in C(b)$ let p_k be the polyhedral selective p-value of Lee et al. [16] for $H_k : \theta_b \le \theta_k$ conditional on E, with the lattice correction of the companion paper. Then $p_{\text{IU}} = \max_{k \in C(b)} p_k$ is super-uniform, conditional on E, under the composite null "some system in $C(b)$ truly dominates b". Rejecting when $p_{\text{IU}} \le \alpha$ certifies at level α that no cheaper-or-equal system is truly at least as accurate as b. That is, b is on the true frontier of its pool.*

World	Rival shape	Tasks	Boards on frontier	Certified at 5% pairwise	composite	exact composite
independent	independent rivals	50	7802	1.1%	2.1%	1.5%
independent	independent rivals	100	8128	1.1%	1.7%	1.1%
independent	independent rivals	150	9259	2.5%	3.3%	2.8%
independent	independent rivals	300	9619	3.6%	4.4%	3.8%
near-copy	near-copy, flip 0.2	50	7757	3.4%	1.6%	0.9%
near-copy	near-copy, flip 0.2	100	8035	5.1%	1.9%	1.3%
near-copy	near-copy, flip 0.2	150	9131	3.3%	3.6%	2.9%
near-copy	near-copy, flip 0.05	300	9661	34.8%	4.2%	1.8%
better	entrant truly better	50	16377	9.8%	14.7%	11.6%
better	entrant truly better	100	18601	23.7%	29.6%	25.0%
better	entrant truly better	150	19319	41.3%	46.5%	42.8%
better	entrant truly better	300	19931	75.9%	78.3%	76.7%

Table 12: What the two certificates control, 20000 boards per row. In the two null worlds, a cheaper independent system with the entrant's true accuracy dominates it. The near-copy rows add a degraded copy of the entrant to the pool. The flip rate is 0.2 at 50 to 150 tasks and 0.05 at 300 tasks. The closer copy is the case the impossibility result is about. In the last block, the entrant is truly better than every cheaper system by half a logit. Rates are over the boards on which the entrant landed on the observed frontier. A valid Pareto certificate must stay at or below 5% in the null rows.

Proof. Under the composite null some H_k holds. E is a polyhedron in $\bar{y}$, so by the polyhedral lemma p_k is uniform conditional on E when $\theta_b = \theta_k$ and super-uniform when $\theta_b < \theta_k$. So $\mathbb{P}(p_{\mathrm{IU}} \le \alpha \mid E) \le \mathbb{P}(p_k \le \alpha \mid E) \le \alpha$. An intersection-union test needs no correction for the number of rivals [6]. □

Two things sit outside the proof. First, the covariance is a plug-in estimate. Second, the lattice correction is a continuity adjustment for accuracies that live on the grid $1/T$. It relaxes each constraint by half a step, and it takes the tail from half a step below the observed value. One of these two actions is conservative, and the other is not. The null rows of Table 12, at most 4.2% and 4.4%, are the check on both. The two certificates answer different questions and have different prices. The pairwise test is exact for any T. It asks whether b beat the system it displaced. The composite test is exact only under normality with a plug-in covariance. It asks whether b beats every cheaper system at once, and it conditions on an event that truncates the statistic more. Table 12 shows both tests on the three simulated worlds. With a near-copy rival, the pairwise test certifies a truly dominated entrant in 3.3 to 34.8% of the cases in which it lands on the frontier. The composite test does so in 1.6 to 4.2%. With independent rivals, the pairwise test certifies 1.1 to 3.6% and the composite test 1.7 to 4.4%. When the entrant is truly better than every cheaper system by half a logit, the pairwise test certifies it 10 to 76% of the time and the composite test 15 to 78%. The composite test gives up nothing in power for its stronger claim. The exact sign test is conservative because its statistic is discrete. This shows in its null rates (1.1 to 3.6% against the composite test's 1.7 to 4.4%). The normal statistic has no such step. What the composite test gives up is exactness. The audit reports both.

Proposition 7 (exact Pareto certificate)**.** *Fix the costs. Outcomes are pass/fail. For $k \in C(b)$ let $\mathcal{G}_k$ be the information in the unordered pairs $\{y_{bt}, y_{kt}\}$ and in the outcomes of all other systems. Let H_k be the statement that tasks are independent and, on every task with $y_{bt} \neq y_{kt}$, $\mathbb{P}(y_{bt} = 1 \mid \mathcal{G}_k) \le \frac{1}{2}$. Equality is the sharp null that b and k are exchangeable given the others. Let D_k be the number of tasks on which b and k differ, N_k the number b wins, n_{11} the number both pass, and*

$$n_{0k} = \max\Big\{\lceil D_k/2 \rceil,\ \lceil T \max_{j \in C(b)\setminus k} \bar{y}_j - n_{11} \rceil\Big\}, \qquad p_k = \frac{\mathbb{P}(X \ge N_k)}{\mathbb{P}(X \ge n_{0k})}, \quad X \sim \mathrm{Bin}(D_k, \tfrac{1}{2}).$$

On the frontier event $E = \{\bar{y}_b \ge \bar{y}_j \text{ for every } j \in C(b)\}$, two things hold. (i) Under H_k, $\mathbb{P}(p_k \le u \mid \mathcal{G}_k, E) \le u$ for every u, with equality at the attainable values under the sharp null. (ii) If H_k holds for some $k \in C(b)$,

then $\mathbb{P}(\max_k p_k \le \alpha \mid E) \le \alpha$*, for any number of tasks. For outcomes in* $[0,1]$ *that are not pass/fail, the same construction by random swaps is valid under the sharp null only.*

Proof. Given $\mathcal{G}_k$, the unordered pair on each discordant task is fixed. So N_k is a sum of D_k independent indicators with success probabilities $q_t \le \frac{1}{2}$ under H_k, and it is $\mathrm{Bin}(D_k, \frac{1}{2})$ under the sharp null. Swapping b's and k's outcomes on a set of tasks leaves $\bar{y}_b + \bar{y}_k$ unchanged. It also leaves every $\bar{y}_j$ with $j \notin \{b, k\}$ unchanged. So given $\mathcal{G}_k$, the event E is the event $\{N_k \ge n_{0k}\}$, with n_{0k} a function of $\mathcal{G}_k$. The first term of the maximum comes from $\bar{y}_b \ge \bar{y}_k$. The second comes from $\bar{y}_b \ge \bar{y}_j$ for the other systems, and it is omitted when $C(b) = \{k\}$. Under the sharp null, the law of N_k given $\mathcal{G}_k$ and E is $\mathrm{Bin}(D_k, \frac{1}{2})$ truncated to $\{N \ge n_{0k}\}$. And p_k is the tail probability of that law at the observed value, so it is exactly uniform on its attainable values. Under H_k, the probability mass function of N_k relative to that of $\mathrm{Bin}(D_k, \frac{1}{2})$ is proportional to $e_n(r)/\binom{D_k}{n}$, with e_n the elementary symmetric polynomial and $r_t = q_t/(1-q_t) \le 1$. Let $S_n = e_n(r)/\binom{D_k}{n}$. Maclaurin's inequality gives $S_{n+1}^{1/(n+1)} \le S_n^{1/n}$. So $S_{n+1} \le S_n \cdot S_n^{1/n} \le S_n$, because $S_n \le 1$ for $r_t \le 1$. The likelihood ratio is therefore nonincreasing. So N_k is smaller than $\mathrm{Bin}(D_k, \frac{1}{2})$ in the likelihood-ratio order, and truncation to $\{N \ge n_{0k}\}$ preserves this order. So $\mathbb{P}(p_k \le u \mid \mathcal{G}_k, E) \le u$, which is (i). For (ii), E is the same event for every k. If H_{k_0} holds, then $\{\max_k p_k \le \alpha\} \subseteq \{p_{k_0} \le \alpha\}$. Averaging (i) over $\mathcal{G}_{k_0}$ given E gives the bound. An intersection-union test needs no correction for the number of rivals [6]. □

The p-value is the sign test of the companion paper with a higher truncation point, $n_{0k} \ge \lceil D_k/2 \rceil$. It needs no normality, covariance estimate, lattice correction or Monte Carlo. It needs only pass/fail outcomes, and that the null be read as conditional on the other systems' outcomes. That last point is the price. A system that truly dominates b in means need not satisfy H_k. In the near-copy world, b wins every discordant task against the independent dominating system on which the copy passes, so $\mathbb{P}(y_{bt} = 1 \mid \mathcal{G}_k) = 1$ there. The proof does not cover that world. Table 12 is the evidence for this. In the audit's three simulated worlds, the exact certificate certifies a dominated entrant 0.9 to 2.9% of the time with a near-copy rival and 1.1 to 3.8% with independent ones. It certifies a truly better entrant 12 to 77% of the time. These rates are on 20000 boards per row. The guarantee is conditional on the weak event E ($\ge$). Certification also requires $\bar{y}_b > \bar{y}_j$ for every j, which can only reduce the probability of certification.

Proposition 8 (no pairwise certificate holds its level)**.** *Let* ϕ_T *be a test of the composite null that* b *is dominated by some cheaper system. The test uses the data only through the outcomes of the entrant* b *and of the observed most accurate cheaper system* j^**. Suppose there is* $\varepsilon(m) \to 0$ *such that* ϕ_T *rejects with probability at least* $1 - \varepsilon(m)$ *whenever* b *wins every task on which* b *and* j^* *differ and there are at least* m *such tasks. Then there are configurations with* $\mu_b \in (0,1)$ *that have three properties. In them,* b *is truly dominated by a strictly cheaper system. The frontier event has probability tending to* $\frac{1}{2}$*. And the conditional rejection probability given the frontier event tends to one.*

Proof. Put in $C(b)$ two systems. One is a strictly cheaper system k with $\mu_k = \mu_b$, independent of b. The other is a cheaper copy j of b whose passes are flipped to failures independently with probability $\delta_T = T^{-3/4}$. Then k dominates b. Let $F \approx \mu_b \delta_T T = \mu_b T^{1/4} \to \infty$ be the number of flipped passes. Then $\bar{y}_b > \bar{y}_j$ with probability tending to one, and $\bar{y}_b > \bar{y}_k$ with probability tending to $\frac{1}{2}$. So $\mathbb{P}(E) \to \frac{1}{2}$. Given E, $j^* = j$ exactly when $\bar{y}_b - \bar{y}_k \ge F/T$. The left side is of order $T^{-1/2}$, with a continuous positive limit density at 0^+, while F/T is of order $T^{-3/4}$. So $\mathbb{P}(j^* = j \mid E) \to 1$. Against j, b wins all F discordant tasks and loses none, and $F \to \infty$ in probability. For each fixed m, $\liminf_T \mathbb{P}(\phi_T \text{ rejects} \mid E) \ge 1 - \varepsilon(m)$, because given E both $F \ge m$ and $j^* = j$ hold with probability tending to one. Letting $m \to \infty$ gives the claim. □

The class is not vacuous. Any test that rejects once b has won a fixed number of discordant tasks and lost none belongs to it, and the selective sign test is one. A test that needs on the order of $\sqrt{T}$ wins before it rejects lies outside it. That is the number an independent rival's sampling noise produces. Such a test pays for its protection by failing to certify a genuine entrant who beats j^* by less than that. The convergence is so slow, of order $T^{-1/4}$, that the proposition describes the shape of the problem and not its size. Table 12 gives the size at 50 to 300 tasks. A leaderboard that lists an agent twice holds a pair of this kind in a weaker form. The two entries may use a stronger and a weaker model, or two reasoning-effort settings. HAL's own reruns of one configuration agree on 72 to 92% of tasks. That is far from the asymptotic construction, which needs agreement on all but about $T^{1/4}$ tasks. The simulations show what pairs that close do at realistic sizes.

So any certificate of Pareto membership must look at the whole cheaper pool. That is what Propositions 6 and 7 do.

Corollary 1 (a chain of entrants)**.** *Let the step index t count the dates on which at least one entrant arrived. Test every entrant of the t-th such date at level α_t (alpha-spending), or at α_t plus the previous such date's level when every entrant of that date was certified (fallback). Multiply its p-value by its per-date factor g. Then the family-wise error of the chain of certificates is at most α. This holds in the sense of Proposition 5 when the pairwise p-values are used, and in the sense of Proposition 6 when the composite ones are.*

Proof. A date's level is a function of which earlier dates had entrants and, for the fallback, of their outcomes. It is a function of no outcome of the date itself. So it is predictable, and the null of each proposition is read conditionally on all earlier systems' outcomes. For a null submission s on date d, $\mathbb{P}(s \text{ certified}) = \mathbb{P}(E_s)\mathbb{P}(p_s \le \alpha_t/g \mid E_s) \le \mathbb{P}(E_s)\,\alpha_t/g$, by Proposition 5 or 6. Sum over the at most g submissions of the date. The expected number of certified null entrants on the date, given the past, is then at most $\alpha_t \mathbb{P}(\text{an entrant arrives} \mid \text{past}) \le \alpha_t$. The step index t_d is predictable and takes distinct values on distinct dates with entrants. So the expected number of certified null entrants over the whole chain is at most $\sum_t \alpha_t = \alpha$. For the fallback, consider the event that date t is the first date on which a null entrant is certified. Every earlier date whose level was passed along the run that feeds t had all its entrants certified, and none of them null. So the run consists of non-null dates. The runs that feed different first-failure dates are disjoint. The levels available to null dates sum to at most α, as in Corollary 2 of Banahene and Opoku [5]. Passing a date's level on when only some of its entrants were certified would break this. A date holding a certified non-null entrant and an uncertified null one would sit inside the run and have its level counted twice. □

Indexing each same-date entrant by its own step would make an entrant's level depend on its rank within the date. That rank is settled only after the date's results are seen. Counting dates avoids this and spends one level per date.

Remark 4 (random costs)**.** *When costs are measured per task, $C(b)$ is itself random. Also, b's mean cost comes from the same run as its outcomes, because tokens are spent on failed tasks too. So the holder is not strictly free of b. For certification, we fix each system's cost at its mean on the day of its arrival. We check the consequence by resampling every system's per-task costs and rebuilding the chain. On the boards with per-task costs, an entrant keeps the same holder in 0.82 to 1.00 of 200 resamples. The number of entrants moves within 50 to 73 around 59. The number certified under alpha-spending moves within 5 to 7 around 5. The frontier set of Section 3.3 is where the cost uncertainty is carried in full.*

B The frontier winner's curse

A system reported as "on the frontier" was selected because it beat every cheaper system. Its reported accuracy is inflated by this selection, like a leaderboard winner's. The selection event, $\bar{y}_b \ge \bar{y}_j$ for all $j \in C(b)$, is a polyhedron in the vector of accuracies. So the conditional method and the iterated bootstrap of Banahene and Opoku [5] apply, with the pool $C(b) \cup \{b\}$ in place of the whole board. The conditional estimate is the median-unbiased estimate under the truncated normal law of $\bar{y}_b$, given that it is the largest of its pool [1, 16]. The bootstrap finds the winner of the pool again in every task resample. It takes the winner's mean excess over its own observed accuracy as the bias. A second bootstrap level removes the bias that the first correction leaves. Costs are fixed at their means, as in certification. A system with no cheaper rival, the cheapest on the board, has a pool of one and nothing to correct.

Table 13 checks both corrections on the simulated boards of Section 5. The reported accuracy of a frontier member overshoots the truth by +0.3 to +2.2 points on average. The larger values come from the boards of 20 systems. The iterated bootstrap leaves a bias of at most 0.8 points, and the conditional estimate leaves at most 0.7. The reported interval covers 0.93 to 0.95. The conditional interval covers 0.95 to 0.96. The one-step bootstrap interval covers 0.90 to 0.94. The iterated bootstrap interval covers 0.89 to 0.93. The two bootstrap corrections remove bias at the cost of coverage, as the companion paper found for the leaderboard winner. Only the conditional estimate improves on both. The frontier curse is milder than the leaderboard winner's curse of the companion paper, because the pool is smaller than the board. It is

				Bias, points (standard error)				Coverage of the 95% interval			
Systems	Tasks	Winners	Pool	reported	bootstrap	iterated	conditional	reported	bootstrap	iterated	conditional
8	50	796	4.6	+1.3 (0.2)	-0.7 (0.2)	-0.8 (0.3)	-0.7 (0.3)	0.95	0.91	0.90	0.96
8	150	841	4.6	+0.8 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.2)	0.94	0.92	0.90	0.95
8	500	836	4.8	+0.3 (0.1)	+0.0 (0.1)	+0.0 (0.1)	-0.2 (0.1)	0.95	0.94	0.93	0.96
20	50	1242	9.3	+2.2 (0.2)	-0.5 (0.2)	-0.7 (0.2)	-0.4 (0.2)	0.93	0.90	0.89	0.96
20	150	1319	9.2	+1.1 (0.1)	+0.2 (0.1)	+0.2 (0.1)	-0.1 (0.1)	0.94	0.92	0.91	0.95
20	500	1328	9.1	+0.3 (0.1)	-0.0 (0.1)	-0.0 (0.1)	-0.3 (0.1)	0.95	0.93	0.92	0.95

Table 13: The frontier winner's curse when the true values are known. Each scenario has 300 boards, and every frontier member with at least one cheaper rival is included. Winners: the number of such members over the 300 boards. Pool: the mean number of systems that cost no more. Bias: the mean of estimate minus true accuracy, in points, with its Monte Carlo standard error. Coverage: the share of 95% intervals that contain the true accuracy.

			Shrinkage, iterated bootstrap (pt)		Shrinkage, conditional (pt)	Below strongest
Benchmark	On frontier	With a cheaper rival	mean	max	mean	cheaper rival after adjustment
assistantbench	3	2	3.3	5.3	1.1	1
colbench backend programming	9	8	0.3	0.5	0.1	0
colbench frontend design	4	3	0.2	0.5	0.4	1
corebench hard	8	7	4.3	7.7	3.5	4
gaia	10	9	1.2	2.4	1.3	2
online mind2web[†]	6	5	0.6	0.9	0.1	0
scicode	3	2	0.2	0.5	0.1	0
scienceagentbench	4	3	0.8	1.6	0.1	0
swebench verified mini	6	5	1.0	2.3	0.1	0
taubench airline	7	6	4.4	9.4	2.8	3
usaco	4	3	0.0	0.1	0.0	0
All	64	53	1.7	9.4	1.1	11

Table 14: Selection-adjusted accuracy of the systems on HAL's observed frontiers. Shrinkage: reported accuracy minus the adjusted estimate, in points, over the frontier members that have a cheaper rival. The last column counts the frontier members whose iterated-bootstrap estimate falls below the reported accuracy of their strongest cheaper rival. Every member is listed in the supplement.

strongest for the expensive systems at the top of the staircase, whose pools are largest. Only 75 to 92% of the frontier members with a rival are truly undominated. This is the plain frontier's error rate seen from the other side.

B.1 On HAL

Table 14 applies the corrections to the 64 systems on HAL's observed frontiers, and 53 of them have a cheaper rival. The iterated bootstrap takes 1.7 points off a frontier member's reported accuracy on average, and 9.4 at most. The conditional estimate takes 1.1 on average. For 11 of the 64 members, on 5 boards, the adjusted accuracy falls below the reported accuracy of the strongest cheaper rival. At its adjusted accuracy, the system would not be on the frontier at all. The corrections are largest where the tasks are few and the frontier members many, CORE-Bench hard and tau-bench airline. They are negligible on USACO and ColBench backend, where the tasks are many.

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